

*Faith, Hope, and Christian Charity: How Religion
Explains Giving when Warm Glow and Impure
Altruism Do Not*

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Abstract

Many charities, especially religious ones, spend significantly to improve poor community members' living standards. Conventional theoretical philanthropy models with altruism or impure altruism/warm glow cannot explain poverty-alleviating private transfers in the type and size of communities where this occurs - e.g., congregations. However, if preferences reflect religious precepts ("you are your brother's keeper"; "treat neighbours like yourself"), charitable poverty alleviation can be optimal even in large communities. Income and price elasticities for giving prove important. Surprisingly, charitable private transfers are less likely when donors care for recipients' living standard instead of the aggregate amount all recipients get.

List (2011, 177) notes: "what we do not know dwarfs what we do know about the economics of charity." This motivates new approaches to study neglected aspects of giving. I analyse charitable poverty alleviation. I use central tenets of Christianity to explain theoretically and numerically, *inter alia*, why: (i) religion is a large recipient of charity; (ii) a major use of the proceeds is poverty alleviation. However - as shown below - leading philanthropy models, e.g. the "Nash contribution" and joint-product/warm glow models, imply that private transfers to reduce poverty are cost-ineffective in large societies if competing with other uses for charitable funds, like on a pure public good. This cost-ineffectiveness is from the second of two large-number effects in voluntary provision of a public good and a private good. The first is the conventional effect leading to free-riding and public good under-provision. The second arises as a transfer directly benefits a sole recipient; but, its opportunity cost increases with the beneficiaries who would have consumed the public good that it could have financed.¹ Warm glow overcomes the first large-number effect but not the second. So, it does not explain giving to alleviate poverty. I show that preferences embodying the central tenets of religion overcome the second effect.

Christianity's central tenets that underpin religious benevolence include: "You are your brother's keeper"; "Love your neighbour like yourself" and "Without charity, I am nothing and I gain nothing." "So faith, hope, and charity abide, these three. But the greatest of these is charity" (1 Corinthians 13:1-13). Thus, unsurprisingly in a deeply pious USA, charitable giving is considerable - the largest portion occurring in religious congregations. Americans give in cash approximately what they spend on clothing, around half their direct spending from post-tax income on healthcare (Bureau of Labour Statistics, 2013). They give religious charities fourteen times what they spend on the 3 major sports (Duin, 2001). Such charities get the largest portion of total US charitable giving and are joint first in the UK. In the US, religion gets circa 35%-40% of total donations by value, nearly thrice the next largest

¹Fraser (1996) shows that this second effect could make it socially optimal to make a poor person worse off by forcing them to contribute to an excludable public good. I think Fraser (2005) is the first model of competing uses of charitable donations for private transfers or a public good where this effect plays a role.

category, education (Andreoni, 2006; Andreoni and Payne, 2013; List, 2011).

Much of that giving is to reduce poverty. Congregations spent \$20bn on philanthropy, 40% of their \$50bn of total US charitable donations of near \$100bn, in 1989 (Biddle, 1992). So, they used 20% of all charitable receipts for philanthropy - expenditures that "address the specific material needs of members of the congregation or [...] support the provision of services to people regardless of their affiliation with the congregation."² Allard (2007) shows how religious charities focus more on material needs than secular charities and, since 1997, have compensated for U.S. governments' reduced stress on cash assistance for the poor.³

This importance of poverty alleviation in motivating charity is not often studied explicitly. Indeed, despite the data, Roberts (1984) claims government welfare spending has crowded out private charity for poverty alleviation. He explicitly analyses agents' spending interpretable as direct monetary transfers to others. While many later crowding out studies exist, there is little focus on this in relation to poverty alleviation.⁴ One exception is Boberg-Fazlic and Sharp (2017). They "find strong empirical support for a positive relationship between welfare provision and charitable giving using data from the late eighteenth and early nineteenth centuries". I.e., they find crowding-in, not crowding-out.

² "60 per cent of all congregations offered a food pantry, 52% offered clothing closets, 44 percent had programs for visiting patients in hospitals, 43 per cent had programs for visiting the sick and 41 per cent sponsored a soup kitchen. These activities aim to help those afflicted by poverty, hunger and disease..." Cnaan and Yancey (2000). Saxon-Harrold et al (2000, p15) state: "In 1997, less than 20% of congregations charged fees for their programs or services. This is far less than other types of non-profit organisations. Nearly 85% of these congregations adjusted their fees by the ability of their recipients to pay..."

³ Allard (2007, p.26) says, "Whether operating a food pantry or meals program, helping a working poor family to make a car payment, or covering utility bills during a particularly expensive winter season, religious non-profit organizations are more likely to offer cash and in-kind assistance than secular nonprofits. For example, faith-based integrated organizations are much more likely to offer food assistance (85 percent respectively) than secular nonprofit organizations (43 percent respectively). Similarly, faith-integrated organizations reported providing emergency cash assistance nearly twice as often as secular nonprofit organizations or government agencies." He quotes Chaves' (2002) finding that below ten percent of congregations provide services outside basic food, clothing or housing needs. Google/Indiana University's Center on Philanthropy (2007) estimates: (i) 84% (\$20.28 bn.) of \$24.14 bn gifted to religious benevolences in 2005 went to the poor; (ii) \$17.03 bn (77%) of Combined Purpose charities' (e.g., United Ways and Jewish Federations) funds did too. Of this, 25.7% met basic needs (e.g., food; housing); (iii) 30.1% of total charitable giving of \$252.55 bn in 2005 directly met the poor's needs.

⁴ de Wit and Bekkers' (2017) extensive meta-analysis of the empirical evidence on crowding out studies how different empirical approaches influence the extent of crowding out (or crowding in) found. They conclude that "private giving is generally not crowded out by public funding". However, they concentrate on possible crowding out in the charitable provision of public goods. Charitable private transfers are not studied.

A facile explanation for a dearth of explicit theoretical analyses of charitable provision of private goods is that, as in the emblematic Bergstrom, Blume and Varian (1986) paper, focus is on "voluntary provision of *public* goods". Authors might not have got to studying charitable provision of private goods with public goods yet. Another argument is Hochman and Rodgers' (1969): "charitable giving, motivated by altruism, creates a public good out of the charity" (Andreoni, 2006). Donors' altruism towards recipients of, e.g., money transfers, daycare or housing makes these, or their institutional arrangements, public goods. But, while the resulting theoretical models consider the public good aspects explicitly, the private goods provided simultaneously, which yield distinct benefits, are not.

The crux of my model is my assumption that the total amount given to alleviate poverty entering agents' utility functions as an argument separate from the amount used for a pure public good embodies religious precepts like being your brother's keeper. This captures List's (2011, 165) idea that different factors to other types of charity motivate religious giving.⁵ He advances social insurance: "it might be [...] that people give to churches so that in time of need they can more readily receive church services or help from fellow churchgoers." Total charitable funding for poverty alleviation being an argument in both donors' and recipients' utility functions reflects satisfaction both from being your brothers' keeper in times of need and knowing that they are yours similarly. I call such preferences "pious". With pious preferences, a utilitarian charity can find poverty alleviating private transfers cost-effective.

My approach complements three empirical strands. One (e.g., Clark and Lelkes, 2006; Scheve and Stasavage, 2006⁶) focuses on religion and redistribution. Religiosity insures against adverse life events for it yields similar psychic benefits as, say, good health, employment, or pension. So, all else equal, the welfare state is smaller as society's religiosity increases as religion's psychic benefits substitute for state benefits.⁷ Another (e.g., Hunger-

⁵ Andreoni and Payne (2013, 10) echo this.

⁶ I am indebted to Roland Benabou for this reference.

⁷ Although not focused on religion and redistribution, Campante and Yanagaziwa-Drott (2015) is relevant to the trade-off between religious satisfaction and material welfare. They show empirically how religious observation (Ramadan fasting) can decrease economic output but increase subjective well-being. More generally, Barro and McCleary (2003) demonstrates that economic growth can increase with religious belief

man, 2005) argues that church welfare spending substitutes directly for the state's. Again, the welfare state is smaller in more religious societies. The third suggests religious giving insures for the afterlife (Hrung, 2004; Chang, 2005; Thornton and Helms, 2013).

My model predicts that, without religion, charitable poverty alleviation will often not occur. So, it rationalises theoretically the first two empirical strands above. People could have preferences of the form that I posit without being explicitly religious. But, if they are religious according to the precepts noted, they will have the preferences that I analyse.

I show that using part of charitable donations for private transfers to the poor can increase giving even if transfers give donors no altruistic externality. An altruistic externality (so transfers function as a public good), can enhance giving, increase welfare and might explain why transfers prevail in typical-sized congregations. Section I reviews theoretical models of giving without religion. Section II shows the difficulty of incorporating poverty alleviation via private transfers in a model of public good provision with standard altruistic preferences. Section III shows how pious preferences overcome this difficulty. Section IV examines other preferences, including joint-product/warm-glow ones. Section V concludes.

I Related Literature: Models of Giving without Religion

The key theoretical papers that I build on are: Becker (1974); Warr (1982, 1983); Cornes and Sandler (1984, 1985, 1994); Roberts (1984); Bergstrom, Blume and Varian (BBV) (1986); Bernheim (1986); Andreoni (1988, 1989, 1990); Fraser (1992, 2005); Vicary (1997) and Ribar and Wilhelm (2002). All, bar Becker, Warr (1982), Roberts and Fraser (2005), study when charitable donations supply a public good without explicit private transfers. Hence, their theoretical models say little about important parts of the charity landscape.⁸

but decline with religious participation. Relatedly, Iannaccone (1992) shows how costly, apparently self-denying, practices can be welfare-improving for religious communities.

⁸An extensive literature builds on Becker (1974) to consider the motives for and consequences of private transfers within households and families. See, e.g., Cox (1987); Cox and Jakubson (1995) and Cox, Hansen and Jimenez (2004). However, like de Wit and Bekkers, I only examine charity where donors do not have a personal relationship with the recipients.

BBV's "classic" Nash equilibrium model gets striking findings via assumption (A.1):

(A.1) Individuals (i 's) have quasi-concave utility functions $u^i[x^i, G]$, $i = 1, \dots, n$, defined over private good consumption, x^i , and public good consumption, G , which are non-decreasing in their arguments. Incomes (m^i), are exogenous.⁹ $G = \sum_i g^i$ sums individual contributions ($g^i \geq 0$) to providing the public good.

These findings include: (i) *the neutrality property* (Warr, 1983): if income is redistributed between donors so that a loser loses no more than his original donation, total donations are unchanged. An extension of this model (Andreoni, 1988) produces (ii) *the limit result*: in large economies, hardly anyone donates.

A consensus exists that this "Nash contribution model" predicts "too little" giving and must be modified. The most relevant modification here is Cornes and Sandler's (1984) and Andreoni's (1989, 1990) "joint product/impure altruism"/"warm glow" model. In this, a donor benefits from both the public good and giving *per se*. So, still using u for a utility function, theirs is $u^i[x^i, g^i, G]$, again quasi-concave non-decreasing. The g^i , a private good for i , gives i an extra spur to donate. This can alter the neutrality and limit results (Andreoni, 1989, 1990; Ribar and Wilhelm, 2002).¹⁰

In my model, all consumers get utility from: (i) total private transfers to the poor; (ii) spending on a shared good like a church building. So, it has two public goods, both subject to crowding-out incentives. However, I show that we can still get total giving and the share of it used for transfers that are "realistic" in a large society. My model relates to Ghosh, Karaivanov and Oak (GKO; 2007) and Cornes and Itaya (2010), who also study multiple public goods. GKO study a charity's incentive to bundle two public goods, not a private and a public good, or private transfers. Their two-consumer model also ignores the size effects that are key in mine. Cornes and Itaya study many public goods and many consumers. They focus mainly on the neutrality result and also ignore charitable transfers.¹¹

⁹Bernheim (1986) considers endogenous incomes.

¹⁰Allouch (2013) elaborates competitive equilibrium in a warm-glow economy. Evren and Minardi (2017) provide an axiomatic foundation for warm-glow preferences.

¹¹Faias et al (2014) extend BBV's results to multiple public and private goods.

Becker (1974), Warr (1982), Roberts (1984) and Brown (1997) are among few who theoretically model charitable private transfers. Becker, Warr and Roberts' donors' utilities depend on the sole recipient's welfare, measured by total income. Total transfers to the poor are then a public good, separate from any altruistic externality from giving. However, Becker's single-donor, single-recipient model has no population effects that decide the value of charitable private transfers. Brown ignores recipients, beyond that the average donation to them is an altruistic externality in donors' utility functions. I show below that if donors care for average rather than total donations, this militates against private transfers. None of these authors considers that intermediaries often organize charity and make transfers if and only if they are cost-effective. This depends on the number of consumers of any public good that funds could finance alternatively, which all ignore by studying a sole use for donations.¹²

Maybe closest to my analysis, Kotchen (2006) models a scenario applicable to charity for a good with public and private characteristics. E.g., buying a grocery item might lead to a fixed donation toward buying a public good. His model's principal differences to mine are twofold: first, its mix of characteristics is fixed exogenously in a one-stage game. In mine, taking the church/charity as bundling a public good and private transfers, the proportions these are combined in arises endogenously from a strategic interaction between charity and consumers in a two-stage game. Second, while he examines how group size affects equilibrium provision of the public characteristic, his model is not set up to show how group size impacts whether or not charitable private transfers are made between consumers.

Despite a dearth of theory on charitable private transfers or religious charities, an impressive empirical religious charity literature exists. Bekkers and Wieping (2011) review key articles. Here, its main significance is to give estimated elasticities to situate my results.

¹²Three more recent analyses of poverty-reducing private transfers are Atkinson (2009), for overseas aid, Echazu and Nocetti (2015), for a discrete good, and Bourlès et al (2017) for altruism in a network. Donations in all three are used solely for a private good. There is no bundling of public and private goods.

II Poverty Alleviation Without Pious Preferences¹³.

Imagine a religious congregation run by "a welfarist charity" intermediary - e.g., Church Elders. Like a Stackelberg leader, in stage 1 of a 2-stage game with congregationers, it collects funds for two mutually-exclusive stage-2 uses: (i) financing services and facilities, including upkeep of the church and pastor, which benefit all members as Samuelson public goods; (ii) private goods and services like food, clothing and shelter, alleviating congregationers' poverty. It creates an altruistic externality if, while not consuming these goods, e.g., soup kitchens, donors get utility as recipients benefit. Giving then creates a public good via this externality. It also provides private goods for soup kitchen patrons - not usually modelled, but part of our focus. We study cases without and with this externality. The charity fixes the portions of total donations used for public good supply and to alleviate poverty, mindful of how giving by congregationers ("followers") depends on them. So, even if donors do not care about poverty alleviation per se, the charity does via its overall welfare concern.

To focus sharply on poverty alleviation, my "two-class model" makes these assumptions:

(A.2) There are two types, indexed h and l , respectively, with exogenous "high" and "low" incomes $m^i > 0$, $i = h, l$, $m^h > m^l$. There are n^i ($n^i \geq 1$) individuals of type i , $i = h, l$.

(A.3) Both types have the concave increasing utility function $U(x, G)$ (i.e., with subscripts for partial derivative, $U_i > 0$, $U_{ii} \leq 0$, $i = x, G$). Respectively, x and G are amounts of the infinitely divisible sole private and sole public good with unit resource costs of unity.

(A.4)(i) The utility function is supermodular: $\frac{\partial^2 U}{\partial x \partial G} \equiv U_{xG} \geq 0$. (ii) Each good is essential and satisfies an Inada condition: $\lim_{x \searrow 0} U_x(x, G) = \infty$, $\lim_{G \searrow 0} U_G(x, G) = \infty$.

In a richer model of more than two income types, a charity would endogenize which qualify for transfers. For tractability, I limit the model to skirt this. Tractability is also a defence of (A.3). Maybe more compelling justifications for (A.3) are: (i) while knowing that taste differences exist, economists rather explain differences in behavior by those in prices and incomes, thus opportunities; (ii) as I focus on behaviour within religious communities,

¹³Some of the steps in this section were first taken in Fraser (2005).

their members might have similar tastes à la Tiebout. If taste differences do exist, well-being is not obviously measured just by income. But, in practice, most charities use income to differentiate "rich" from "poor". (A.3) means that, initially, I assume consumers get no altruistic externality from any private transfers to the poor.

(A.4)(i) ("Edgeworth-Pareto complementarity" of x and G) is needed for monotone comparative statics. With only two goods, x and G are Hicksian substitutes. (A.4)(ii) implies that any equilibrium is interior.

The Charity

At stage 2, the utilitarian charity¹⁴, sole provider of the tangible public good, uses donations to supply it and for transfers to the poor l types in proportions it fixes at stage 1 to reflect its concern for inequality. I focus first on stage 2 under the next assumption:

(A.5) The charity uses a share $\lambda \in [0, 1]$ of its receipts to supply the public good and $1 - \lambda$ for the private transfers. It can identify the poor and divides transfers equally: each recipient gets $(1 - \lambda) / n^l$ from a total transfer of $1 - \lambda$.

(A.5) implies that, as λ decreases, it costs more for an h type to augment G by giving. Simultaneously, it is cheaper for an l type, if h types are also giving.

Consumer Behaviour

Consider a given regime defined by the charity's chosen λ . Suppose an arbitrary l -type agent conjectures that the $n^l - 1$ others each gives $\bar{g}^l$ to charity and the n^h h types each gives $\bar{g}^h$. Then, if this l type donates g^l , it conjectures that it will receive a transfer, c^l , satisfying $n^l c^l = (1 - \lambda) [n^h \bar{g}^h + (n^l - 1) \bar{g}^l + g^l] = (1 - \lambda) (G + n^l c^l)$, i.e.,

$$c^l = \left(\frac{1 - \lambda}{n^l} \right) [n^h \bar{g}^h + (n^l - 1) \bar{g}^l + g^l] = \left(\frac{1 - \lambda}{\lambda n^l} \right) G \quad (1)$$

¹⁴Bilodeau and Slivinski (1997) study charities' motivations and competition. Instead of utilitarianism, we could consider a charity that allocates funds democratically. Roberts (1984) uses a median voter model and Scharf (2000) a "citizen-candidate" one to show how democracy might shape the environment for giving.

Hence, by the first equality in (1), the l type has the budget constraint

$$x^l = m^l + c^l - g^l = m^l + \left(\frac{1-\lambda}{n^l} \right) [n^h \bar{g}^h + (n^l - 1) \bar{g}^l] - \left(\frac{n^l + \lambda - 1}{n^l} \right) g^l \quad (2)$$

If an h type conjectures that each of the other $n^h - 1$ of its type gives $\bar{g}^h$ and the n^l of type l each gives $\bar{g}^l$, as it gets no transfer, its budget constraint is

$$x^h = m^h - g^h \quad (3)$$

Let $G_{-l} \equiv \lambda [n^h \bar{g}^h + (n^l - 1) \bar{g}^l]$ denote others' total contribution to the public good supply conjectured by an arbitrary type l . Then, by

$$G = G_{-l} + \lambda g^l \implies g^l = \frac{(G - G_{-l})}{\lambda} \quad (4)$$

the outer equality in (1) and the first in (2), its rewritten budget constraint is

$$m^l + \frac{G_{-l}}{\lambda} = x^l + \left[\frac{(n^l + \lambda - 1)}{\lambda n^l} \right] G \quad (5)$$

The LHS of (5) is an arbitrary type l 's "full income", given others' donations. With this income, it chooses its desired G at an effective price $\frac{(n^l + \lambda - 1)}{\lambda n^l}$ per unit.

Defining G_{-h} similarly, an h type's budget constraint is

$$m^h + \frac{G_{-h}}{\lambda} = x^h + \left(\frac{G}{\lambda} \right) \quad (6)$$

By (5)-(6), G 's effective price for an h (i.e., $\frac{1}{\lambda}$) exceeds that for an l (i.e., $\frac{(n^l + \lambda - 1)}{\lambda n^l}$) if $\lambda < 1$. So, the poor might give to charity despite receiving charitable transfers as they find giving relatively cheap. However, their incentive to donate falls rapidly as n^l increases.

Using (5)-(6), the types' maximization problems are

$$Max_{G \geq G_{-h}}.U \left[m^h + \frac{G_{-h}}{\lambda} - \frac{G}{\lambda}, G \right]; \quad (7)$$

$$Max_{G \geq G_{-l}}.U \left[m^l + \frac{G_{-l}}{\lambda} - \left(\frac{n^l + \lambda - 1}{\lambda n^l} \right) G, G \right] \quad (8)$$

Analogous to Bergstrom, Blume and Varian (1986), these yield public good demand functions $f \left[m^h + \frac{G_{-h}}{\lambda}, \frac{1}{\lambda} \right]$ and $f \left[m^l + \frac{G_{-l}}{\lambda}, \left(\frac{n^l + \lambda - 1}{\lambda n^l} \right) \right]$ satisfying $G \geq f \left[m^h + \frac{G_{-h}}{\lambda}, \frac{1}{\lambda} \right]; G \geq f \left[m^l + \frac{G_{-l}}{\lambda}, \left(\frac{n^l + \lambda - 1}{\lambda n^l} \right) \right]$ for some function $f(.)$. As G is essential, at least one of these weak inequalities is an equality as a $G > 0$ must be the optimal demand for at least one type; if both types donate, both are equalities. At such an interior equilibrium, private consumption and utilities are not equalized, unlike in the standard Nash model. If $\lambda < 1$, the l type faces a lower effective price for G , yet demands the same as the h . So, with G normal, the l type has lower equilibrium income and private consumption.

The second-stage equilibrium

I study a subgame perfect equilibrium. At stage 2, this involves a Nash equilibrium in giving. I study a symmetric equilibrium with all members of an income type behaving identically. For a given λ , this stage-2 equilibrium can be shown to exist and be unique.

Let us use the following notation for the arbitrary utilities of the two types:

$$L(g^h, g^l, \lambda) \equiv U \left[m^l + \left(\frac{1 - \lambda}{n^l} \right) (n^h \bar{g}^h + (n^l - 1) \bar{g}^l) + \left(\frac{1 - \lambda - n^l}{n^l} \right) g^l, G_{-l} + \lambda g^l \right]; \quad (9)$$

$$H(g^h, g^l, \lambda) \equiv U [m^h - g^h, G_{-h} + \lambda g^h] \quad (10)$$

Let $U^i[.]$ be a utility magnitude for an $i, i = h, l$, with omitted functional arguments. Then, with each person taking others' donations as given, using subscripts to show partial deriva-

tives for the indicated variables, their donations in a Nash equilibrium satisfy, respectively,

$$L_{g^l} = \left(\frac{1 - \lambda - n^l}{n^l} \right) U_x^l [\cdot] + \lambda U_G^l [\cdot] \leq 0, g^l \geq 0; \quad (11)$$

$$H_{g^h} = -U_x^h [\cdot] + \lambda U_G^h [\cdot] \leq 0, g^h \geq 0 \quad (12)$$

By Kuhn-Tucker theory, the inequalities in (11) cannot both be strict, likewise those in (12). Also, as the public good is essential, at least one of $g^h > 0$ or $g^l > 0$ holds, so one of L_{g^l} and H_{g^h} equals zero.

The Charity's Maximisation

The utilitarian charity's maximand (i.e., social welfare) is

$$W \equiv n^h H(g^h, g^l, \lambda) + n^l L(g^h, g^l, \lambda) \quad (13)$$

Intuitively obviously, with an income distribution such that both types wish to contribute at $\lambda = 1$, there are no poverty-alleviating transfers. Incomes are equalized in the Nash equilibrium and $\frac{\partial W}{\partial \lambda}|_{\lambda=1} \geq 0$ (Fraser, 2005). Relatedly, Fraser (2005) shows, again intuitively obviously, that an utilitarian charity equilibrium never has the poor donating. So, here, I study scenarios with income distributions such that only the rich donate when $\lambda = 1$ and explore when the charity wishes to alleviate poverty then. This must be social welfare improving. Thus, we need $\frac{\partial W}{\partial \lambda}|_{\lambda=1} < 0$. I.e., if welfare is decreasing in λ at $\lambda = 1$, then reducing λ from $\lambda = 1$ increases welfare over some range of λ .

Denote $g^i(\lambda)$, $g^{i'}(\lambda)$, $i = h, l$, as the levels and derivatives with respect to λ of types' donations at an arbitrary λ and use $g^l(1) = 0 = g^{l'}(1)$. Then (omitting functional arguments)

$$\begin{aligned} \frac{\partial W}{\partial \lambda}|_{\lambda=1} &= n^h U_G^h [n^h g^h + (n^h - 1) g^{h'}] + n^l \left[U_x^l \left(-\frac{n^h g^h}{n^l} \right) + U_G^l n^h (g^h + g^{h'}) \right] \\ &= \underbrace{[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h}_{\text{"an income effect"}} + \underbrace{[n^h U_G^h + n^l U_G^l - U_x^h] n^h g^{h'}}_{\text{"a price effect"}} \end{aligned} \quad (14)$$

$$= \underbrace{[n^h U_G^h + n^l U_G^l] n^h (g^h + g^{h'}) - n^h (g^h U_x^l + g^{h'} U_x^h)}_{\text{welfare effect of marginally introducing private transfers}} \quad (15)$$

using $U_G^h = U_x^h$ at $\lambda = 1$ from (12). Also, from differentiating $H_{g^h} = 0$ at $\lambda = 1$,

$g^{h'}(1) = [U_{xx}^h - (n^h + 1) U_{xG}^h + n^h U_{GG}^h]^{-1} [n^h g^h U_{xG}^h - n^h g^h U_{GG}^h - U_G^h]$. So, $U_{xG}^h \geq 0$ ((A.4)(i)) and $-G \frac{U_{GG}^h}{U_G^h} \geq 1$, with one inequality strict, suffice for $g^{h'}(1) < 0$; similarly, an additively separable $U[\cdot]$ and $-G \frac{U_{GG}^h}{U_G^h} < 1$ give $g^{h'}(1) > 0$. We easily find examples with $g^l(1) = 0 = g^{h'}(1)$ and $g^{h'}(1) < 0$ and, so, the charity could increase aggregate giving by making private transfers to the poor (i.e., reducing λ from $\lambda = 1$).¹⁵ The crucial question is, is this welfare-improving in reasonably-sized giving communities like congregations? For this, W must increase as λ decreases from 1: $\frac{\partial W}{\partial \lambda}|_{\lambda=1} < 0$ must hold.¹⁶

In (15), marginally introducing transfers to the poor affects public and private good consumption. All else equal, marginally reducing λ from $\lambda = 1$ reduces G at rate $n^h g^h$, changing giving by $n^h g^{h'}$. To first-order, G reduces by $n^h (g^h + g^{h'})$, of utility value $-[n^h U_G^h + n^l U_G^l] n^h (g^h + g^{h'})$. This yields funds to transfer at a rate $n^h g^h$, raising the poor's utility by $n^h g^h U_x^l$. As reducing λ induces the rich to change giving by $n^h g^{h'}$, their utility from private consumption alters by $n^h g^{h'} U_x^h$. So, (15) nets these utility effects for an *increase* in λ .

Alternatively, (14) shows how community size determines if private transfers are desirable. The "income effect" is the opportunity cost of reducing λ from $\lambda = 1$ and using funds released from G for transfers, all else equal. The reduced λ induces a change in giving of $-n^h g^{h'}(1)$. To first order, the "price effect" is the welfare effect of this altered private giving, net of its impact on donors' private consumption.

Note that $g^{h'}(1) < 0$ is neither necessary nor suffices for $\frac{\partial W}{\partial \lambda}|_{\lambda=1} < 0$: if $g^{h'}(1) \geq 0$, $\frac{\partial W}{\partial \lambda}|_{\lambda=1} < 0$ is possible if $[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h < 0$ at $\lambda = 1$. However, it is immediately evident that the latter is unlikely for large n^h and/or large n^l , as Proposition 1 shows.

PROPOSITION 1. $[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h > 0$ at $\lambda = 1$ if n^h and/or n^l is sufficiently large.

¹⁵ As I show presently, it is also easy to find examples where $g^{h'} < 0$ does not hold.

¹⁶ Andreoni and Payne (2013) cautions that "we must resist the inference that anything that increases donations is good for society."

PROOF. Fraser (2005) shows Nash equilibrium existence and uniqueness in this model for an arbitrary λ . In a symmetric Nash equilibrium for an arbitrary n^h with only the h type contributing at $\lambda = 1$, the optimal g^h satisfies $U_x [m^h - g^h, n^h g^h] = U_G [m^h - g^h, n^h g^h]$ and $U_x [m^l, n^h g^h] \geq U_G [m^l, n^h g^h]$. Now, the optimal g^h is a function of n^h and m^h but is independent of n^l and m^l . So, both $U_x [m^l, n^h g^h]$ and $U_G [m^l, n^h g^h]$ are fixed and finite at a given n^h . Hence, for sufficiently large $n^l = \hat{n}^l > 1$, we have $\hat{n}^l U_G [m^l, n^h g^h] \geq U_x [m^l, n^h g^h]$. Thus, $[n^h U_G (m^h - g^h, n^h g^h) + n^l U_G (m^l, n^h g^h) - U_x (m^l, n^h g^h)] n^h g^h > 0$ for $n^l \geq \hat{n}^l$. Also, $U_G (m^h - g^h, n^h g^h) \geq U_G (m^l, n^h g^h)$ by $U_{xG} \geq 0$. So, for $n^h, n^l \geq \hat{n}^l$, $[n^h U_G (m^h - g^h, n^h g^h) + n^l U_G (m^l, n^h g^h) - U_x (m^l, n^h g^h)] n^h g^h > 0$. ■

Whether $[n^h U_G^h + n^l U_G^l - U_x^l] n^h g^h > 0$ for an actual population (n^h, n^l) can be verified once the utility function is specified.

Next, to closely study what determines $g^{h'}(\lambda)$, I use the properties of the public good demand function, $G \equiv f \left[m^h + \frac{G_{-h}}{\lambda}, \frac{1}{\lambda} \right] = \lambda n^h g^h(\lambda)$. The next proposition relates $sign(g^{h'}(\lambda))$ to the price and income elasticities of giving.

PROPOSITION 2. Let f_m and f_p , respectively, denote the derivatives of $f(\cdot)$ with respect to an h 's full income, $m \equiv m^h + (n^h - 1)g^h(\lambda)$ in a symmetric equilibrium and the public good's effective price, $p \equiv 1/\lambda$. Let $\varepsilon_m \equiv m f_m / f$ and $\varepsilon_p \equiv \lambda^{-1} f_p / f$, respectively, be the income and price elasticity of demand for the public good. Then, for "large" n^h , $\varepsilon_p > -1$ and $\varepsilon_m \leq 1$ suffice but are unnecessary for $g^{h'}(\lambda) < 0$.

PROOF. As $g^h(\lambda) = \frac{f[m^h + (n^h - 1)g^h(\lambda), \frac{1}{\lambda}]}{\lambda n^h}$ in a symmetric Nash equilibrium with only the rich donating, differentiation and manipulation yield

$$g^{h'}(\lambda) = - \left(\frac{f}{\lambda^2 n^h} \right) + \left(\frac{1}{\lambda n^h} \right) \left[f_m (n^h - 1) g^{h'}(\lambda) - f_p \left(\frac{1}{\lambda^2} \right) \right]$$

So

$$g^{h'}(\lambda) \left[1 - \frac{f_m (n^h - 1)}{\lambda n^h} \right] = - \left(\frac{1}{\lambda^2 n^h} \right) \left[f + \left(\frac{1}{\lambda} \right) f_p \right]$$

$$\begin{aligned} \Rightarrow g^{h'}(\lambda) &= -\frac{\left(\frac{1}{\lambda^2 n^h}\right) \left[f + \left(\frac{1}{\lambda}\right) f_p\right]}{\left[1 - f_m (n^h - 1) / \lambda n^h\right]} \\ &\cong -\left(\frac{1}{\lambda^2 n^h}\right) \left[\frac{(1 + \varepsilon_p)}{\left(\frac{1}{f} - \frac{1}{\lambda} \frac{\varepsilon_m}{m}\right)}\right] \end{aligned} \quad (16)$$

on dividing both numerator and denominator by f , using the definitions of ε_m and ε_p , and $\frac{(n^h-1)}{n^h} \cong 1$ for large n^h . If numerator and denominator of the last square-bracketed term have opposite (the same) signs, $g^{h'}(\lambda) > (<)0$. Now, $\left(\frac{1}{f} - \frac{1}{\lambda} \frac{\varepsilon_m}{m}\right) \left\{ \begin{smallmatrix} \geq \\ < \end{smallmatrix} \right\} 0$ as $\frac{\lambda m}{f} \left\{ \begin{smallmatrix} \geq \\ < \end{smallmatrix} \right\} \varepsilon_m$. But $\lambda m = \lambda [m^h + (n^h - 1) g^h] > \lambda n^h g^h = f = G$, as $m^h > g^h$. So, $f^{-1} \lambda m > 1$. Thus, $\varepsilon_m \leq 1$ suffices for $\left(\frac{1}{f} - \frac{1}{\lambda} \frac{\varepsilon_m}{m}\right) > 0$, but $\left(\frac{1}{f} - \frac{1}{\lambda} \frac{\varepsilon_m}{m}\right) > 0$ is possible with $\varepsilon_m > 1$. Also, $(1 + \varepsilon_p) \left\{ \begin{smallmatrix} \geq \\ < \end{smallmatrix} \right\} 0$ as $\varepsilon_p \left\{ \begin{smallmatrix} \geq \\ < \end{smallmatrix} \right\} -1$. Hence, $\varepsilon_p > -1$ and $\varepsilon_m \leq 1$ suffice but are unnecessary for $g^{h'}(\lambda) < 0$. ■

No consensus exists on ε_p and ε_m for charitable giving, for religion or generally, within a large empirical literature. Taussig's (1967) seminal paper finds very small ε_p of the order - 0.04 to - 0.10 and ε_m significantly above 1.0. Later studies - e.g., Auten, Sieg and Clotfelter (2002) - find larger price elasticities but smaller income ones. Their preferred estimates for transitory price elasticities range from - 0.40 to - 0.61 and those for persistent prices from - 0.79 to -1.26. Their estimates of persistent income elasticities range from 0.40 to 0.87 and those for transitory income from 0.29 to 0.45. Karlan and List (2007) argue, based on a large-scale natural field experiment, that $0 \geq \varepsilon_p \geq -1$.

Feldstein (1975) and others (e.g., Backus, 2010, Scharf and Smith, 2010) also disaggregate the data for elasticities of different types of giving. He finds that, in absolute value, the price elasticity is below 1 for religion but exceeds 1 for education, hospitals, health and welfare, and all else. His estimated income elasticities exceed 1 for hospitals and educational institutions but are below 1 for all else. These differences suggests that we could have different signs empirically for $g^{h'}(\lambda)$ for different categories of charitable giving. A tendency for $0 < \varepsilon_m < 1$ and $-1 < \varepsilon_p < 0$ empirically for religious giving implies that $g^{h'}(\lambda) < 0$ might hold with this if either the model of this section or the next one applies.

Lastly, for more insight, I consider a log Cobb-Douglas example, as in Ribar and Wilhelm

(2002) and Andreoni (2006), but without warm glow. So, suppose preferences are $U[x, G] = \alpha_1 \log x + \alpha_2 \log G$, $\alpha_i \in (0, 1)$, $i = 1, 2$, being elasticity of marginal utility parameters.

A symmetric Nash equilibrium with only h persons giving at $\lambda = 1$ satisfies

$$\begin{aligned} U_G(m^h - g^h(1), n^h g^h(1)) &= U_x(m^h - g^h(1), n^h g^h(1)) \quad (a) \\ U_G(m^l, n^h g^h(1)) &\leq U_x(m^l, n^h g^h(1)) \quad (b) \end{aligned} \quad (17)$$

With the log Cobb-Douglas preferences, (17)(a) implies $\frac{\alpha_1}{m^h - g^h(1)} = \frac{\alpha_2}{n^h g^h(1)}$, or $g^h(1) = \alpha_2 m^h / [\alpha_2 + \alpha_1 n^h]$. (17)(b) then implies

$$\frac{\alpha_1}{m^l} \geq \frac{\alpha_2}{n^h g^h(1)} = \frac{\alpha_2 + \alpha_1 n^h}{n^h m^h} \implies \frac{m^h}{m^l} \geq \frac{\alpha_2 + \alpha_1 n^h}{\alpha_1 n^h} = 1 + \frac{\alpha_2}{\alpha_1 n^h}. \quad (18)$$

So, (18) reflects the well-known fact that, in a large community, the income disparity need not be great for only the highest income type to donate.

In this example, $U_{xG}^h = 0 = GU_{GG}^h + U_G^h$, so $g^{h'}(1) = 0$.¹⁷ Thus, from (14),

$$\begin{aligned} \text{sign}\left(\frac{\partial W}{\partial \lambda}\bigg|_{\lambda=1}\right) &= \text{sign}\left[(n^h U_G^h + n^l U_G^l - U_x^l)\big|_{\lambda=1}\right]. \text{ As } g^h(1) = \frac{\alpha_2 m^h}{\alpha_2 + \alpha_1 n^h}, \\ (n^h U_G^h + n^l U_G^l - U_x^l)\big|_{\lambda=1} &= (n^h + n^l) \alpha_2 \frac{(\alpha_2 + \alpha_1 n^h)}{\alpha_2 m^h n^h} - \frac{\alpha_1}{m^l} \\ &\quad \left\{ \begin{array}{l} \geq \\ \leq \end{array} \right\} 0 \text{ as } (n^h + n^l) (\alpha_2 + \alpha_1 n^h) m^l - \alpha_1 m^h n^h \\ &= \alpha_1 n^h [(n^h + n^l) m^l - m^h] + \alpha_2 (n^h + n^l) m^l \left\{ \begin{array}{l} \geq \\ \leq \end{array} \right\} 0. \end{aligned}$$

For an U.S. congregation of mean size 186 or median size 75, $(n^h + n^l) m^l > m^h$: the ratio of the income of a typical giver to that of a typical transfer recipient is much smaller than the median or mean congregation size. So, as this is sufficient, not necessary, for $\frac{\partial W}{\partial \lambda}\big|_{\lambda=1} > 0$, we expect $\frac{\partial W}{\partial \lambda}\big|_{\lambda=1} > 0$, irrespective of the parameters $\alpha_i > 0$. Thus we expect no private transfers to occur with log Cobb-Douglas utility.

¹⁷Note that $g^{h'}(1) = 0$ is consistent with the approximate formula for $g^{h'}(\lambda)$ in (16) above. With Cobb-Douglas utility, the price elasticity of demand for the public good is -1 , so the numerator of the $g^{h'}(\lambda)$ expression in (16) equals 0. To see this in the current context, note that when each h person maintains g^h as λ changes, consumption of $G = \lambda n^h g^h$ must change exactly in proportion - i.e., the price elasticity $= -1$ as the effective unit price of G is λ^{-1} .

III Poverty Alleviation with Pious Preferences

To reflect the assumption that all congregationers care about the total amount of funds being transferred to the poor, denoted T , I now posit that preferences satisfy (A.6):

(A.6). All households have the utility function $U[x, T, G]$, x and G defined as before. U is concave increasing in its arguments and satisfy Inada conditions for x and G .

(A.6) is consistent with Hochman and Rodgers' conception of interdependent preferences, with the interdependence as an external economy. I assume that transfers, T , do not satisfy an Inada condition for two reasons. First, to do otherwise forces the model to generate the very behaviour that I seek to explain. I do not want assumptions on preferences to force transfers to the poor to be made; I want them to occur if the charity finds it cost-effective to make them. Second, at the model's highly aggregative level, private consumption of some sort is indispensable to every consumer, as is public good consumption. However, giving to the poor need not be perceived as indispensable to anyone.

A charity that behaves as analyzed above and transfers a fraction $1 - \lambda$ of total donations to the poor will, in a symmetric equilibrium, make total transfers of

$$T = (1 - \lambda) \left[\sum_{j=1}^{n^h} g_j^h + \sum_{k=1}^{n^l} g_k^l \right] = (1 - \lambda) [n^h \bar{g}^h + n^l \bar{g}^l] \quad (19)$$

Using $G = \lambda(T + G)$, $T = (1 - \lambda)(T + G) = \left(\frac{1-\lambda}{\lambda}\right) G$, an arbitrary poor person of pious preferences now maximises $U[x^l, T, G] = U[x^l, \left(\frac{1-\lambda}{\lambda}\right) G, G]$ subject to the budget constraint

$$\begin{aligned} x^l + g_l &\leq m^l + (1 - \lambda) n^{l-1} [n^h \bar{g}^h + (n^l - 1) \bar{g}^l + g^l] \\ \implies x^l + \lambda g_l + (1 - \lambda) g_l + \lambda [n^h \bar{g}^h + (n^l - 1) \bar{g}^l] + (1 - \lambda) [n^h \bar{g}^h + (n^l - 1) \bar{g}^l] \\ &\leq m^l + (1 - \lambda) n^{l-1} [n^h \bar{g}^h + (n^l - 1) \bar{g}^l + g^l] + \lambda [n^h \bar{g}^h + (n^l - 1) \bar{g}^l] + (1 - \lambda) [n^h \bar{g}^h + (n^l - 1) \bar{g}^l] \\ \implies x^l + T + G &\leq m^l + (1 - \lambda) n^{l-1} [T + G] + T_{-l} + G_{-l} \\ \implies x^l + (T + G) \left[\frac{n^l + \lambda - 1}{n^l} \right] &\leq m^l + T_{-l} + G_{-l} \implies x^l + \left[\frac{n^l + \lambda - 1}{n^l} \right] \frac{G}{\lambda} \leq m^l + \frac{G_{-l}}{\lambda}. \end{aligned}$$

So, with non-satiation, this person has the optimisation

$$\begin{aligned} \underset{x^l; G \geq G_{-l}}{Max.} \ U \left[x^l, \left(\frac{1-\lambda}{\lambda} \right) G, G \right] \text{ s.t. } x^l + \left[\frac{n^l + \lambda - 1}{n^l} \right] \frac{G}{\lambda} = m^l + \frac{G_{-l}}{\lambda} \\ \iff \underset{G \geq G_{-l}}{Max.} U \left[m^l + \frac{G_{-l}}{\lambda} - \left[\frac{n^l + \lambda - 1}{n^l} \right] \frac{G}{\lambda}, \left(\frac{1-\lambda}{\lambda} \right) G, G \right] \end{aligned} \quad (20)$$

The analogous optimisation for an the arbitrary rich person is

$$\begin{aligned} \underset{x^h; G \geq G_{-h}}{Max.} \ U \left[x^h, \left(\frac{1-\lambda}{\lambda} \right) G, G \right] \text{ s.t. } x^h + \frac{G}{\lambda} = m^h + \frac{G_{-h}}{\lambda} \\ \iff \underset{G \geq G_{-h}}{Max.} U \left[m^h + \frac{G_{-h}}{\lambda} - G, \left(\frac{1-\lambda}{\lambda} \right) G, G \right] \end{aligned} \quad (21)$$

The crucial difference from optimising with non-pious preferences is the new argument in utility, $T = \left(\frac{1-\lambda}{\lambda} \right) G$. This yields public good demands $G^p = f^{ph} \left[m^h + \frac{G_{-h}}{\lambda}, \left(\frac{1-\lambda}{\lambda} \right) G_{-h}, \frac{1}{\lambda} \right]$ and $G^p = f^{pl} \left[m^l + \frac{G_{-l}}{\lambda}, \left(\frac{1-\lambda}{\lambda} \right) G_{-l}, \frac{(n^l + \lambda - 1)}{\lambda n^l} \right]$, superscript p denoting "from pious preferences."

As before, as only the rich give in an utilitarian charity equilibrium, I examine just their behaviour. Now, $g^{ph}(\lambda) = \frac{f^{ph} \left[m^h + \frac{G_{-h}}{\lambda}, \left(\frac{1-\lambda}{\lambda} \right) G_{-h}, \frac{1}{\lambda} \right]}{\lambda n^h}$. So, differentiating with respect to λ , rearranging, using the same notation as with non-pious preferences, plus $\varepsilon_{rel} \equiv \frac{T_{-h} f_{T_{-h}}^{ph}}{f^{ph}}$, the elasticity of giving due to a concern about the welfare of others, gives

$$g^{ph'}(\lambda) \cong - \left(\frac{1}{\lambda^2 n^h} \right) \left[\frac{\left(1 + \varepsilon_p^p + f_{rel}^{ph} \right)}{\left(\frac{1}{f^{ph}} - \frac{1}{\lambda} \frac{\varepsilon_m^p}{m} - \frac{\varepsilon_{rel}}{G_{-h}} \right)} \right] \quad (22)$$

Compared with non-pious preferences, the *ceteris paribus* impact of pious preferences on $sign(g^{ph'}(\lambda))$ is ambiguous *a priori*. It is influenced by $sign\left(f_{rel}^{ph}\right)$, thus, the identically-signed ε_{rel} . Similar to before, $g^{ph'}(\lambda) > (<)0$ if the numerator and denominator of the square-bracketed term in (22) have opposite (the same) signs. As my exemplary public good is a church, I think reasonably that the religious regard it as complementary with transfers to the poor, whence $f_{rel}^{ph} > 0, \varepsilon_{rel} > 0$. Empirical evidence from Feldstein (1975) on also tends to suggest that giving to religion is less price elastic than non-religious giving, whence $1 + \varepsilon_p^p + f_{rel}^{ph} > 1 + \varepsilon_p^p > 1 + \varepsilon_p > 0$ holds. All else equal, religiosity then could make $g^{ph'}(\lambda)$

more negative if it makes $\left(1 + \varepsilon_p^p + f_{rel}^{ph}\right)(> 0)$ larger than the counterpart $(1 + \varepsilon_p) > 0$ in the non-pious case, and $\frac{1}{f^{ph}} - \frac{1}{\lambda} \frac{\varepsilon_m^p}{m} - \frac{\varepsilon_{rel}}{G_{-h}}(> 0)$ smaller than its non-pious counterpart, $\frac{1}{f^{ph}} - \frac{1}{\lambda} \frac{\varepsilon_m}{m}(> 0)$.¹⁸

Standard comparative statics gives Proposition 3 for $sign(g^{ph'}(\lambda))$:

PROPOSITION 3. $g^{ph'}(1)$ is a priori ambiguous in general. If $U(x, G, T)$ is additively separable, $-G \frac{U_{GG}}{U_G} \geq 1$ suffices for $g^{ph'}(1) < 0$. If $U(x, G, T)$ is log Cobb-Douglas, then $g^{ph'}(1) < 0$.

PROOF. Consider a neighbourhood of $\lambda = 1$ where $g^{ph}(\lambda) > 0 = g^l(\lambda)$. The relevant FOC for an arbitrary h person's $g^h(\lambda)$ at a symmetric equilibrium is

$$\begin{aligned} & U_x [m^h - g^{ph}(\lambda), \lambda n^h g^{ph}(\lambda), (1 - \lambda) n^h g^{ph}(\lambda)] \\ & + \lambda U_G [m^h - g^{ph}(\lambda), \lambda n^h g^{ph}(\lambda), (1 - \lambda) n^h g^{ph}(\lambda)] \\ & - (1 - \lambda) U_T [m^h - g^{ph}(\lambda), \lambda n^h g^{ph}(\lambda), (1 - \lambda) n^h g^{ph}(\lambda)] = 0 \end{aligned} \quad (23)$$

Differentiating (23) with respect to λ , omitting functional arguments,

$$\begin{aligned} 0 = & -U_{xG} (n^h g^{ph} + \lambda n^h g^{ph'}) - U_T + \\ & + (1 - \lambda) [-g^{ph'} U_{xT} + U_{TT} ((1 - \lambda) n^h g^{ph'} - n^h g^{ph})] \\ & + U_{GT} (n^h g^{ph} + \lambda n^h g^{ph'}) + U_G \\ & + \lambda [U_{xG} (-g^{ph'}) + U_{GT} ((1 - \lambda) n^h g^{ph'} - n^h g^{ph}) + U_{GG} (n^h g^{ph} + \lambda n^h g^{ph'})] \end{aligned} \quad (24)$$

$$\implies g^{ph'}(1) = \frac{[U_T - U_G - n^h g^{ph} (U_{xT} - U_{xG} - U_{GT} + U_{GG})]}{[U_{xx} - (n^h + 1) U_{xG} + n^h U_{GG}]} \quad (25)$$

By concavity and supermodularity, $[U_{xx} - (n^h + 1) U_{xG} + n^h U_{GG}] < 0$, but $g^{ph'}(1)$ is a priori ambiguous. If $U(x, G, T)$ is separable, $U_{xT} = U_{xG} = U_{GT} = 0$, and $-G \frac{U_{GG}}{U_G} \geq 1$ suffices for $g^{ph'}(1) < 0$. When $U(x, G, T)$ is log Cobb-Douglas, $-G \frac{U_{GG}}{U_G} = 1$. ■

¹⁸But, other combinations are possible. E.g., $f_{rel}^{ph} > 0$ might produce: (i) $(1 + \varepsilon_p^p + f_{rel}^{ph}) > 0$ if $(1 + \varepsilon_p) < 0$, and $\frac{1}{f^{ph}} - \frac{1}{\lambda} \frac{\varepsilon_m^p}{m} - \frac{\varepsilon_{rel}}{G_{-h}} < 0$ if $\frac{1}{f} - \frac{1}{\lambda} \frac{\varepsilon_m}{m} < 0$ in the non-pious case; (ii) $(1 + \varepsilon_p^p + f_{rel}^{ph}) > 0$ if $(1 + \varepsilon_p) > 0$ and $\frac{1}{f^{ph}} - \frac{1}{\lambda} \frac{\varepsilon_m^p}{m} - \frac{\varepsilon_{rel}}{G_{-h}} < 0$ if $\frac{1}{f} - \frac{1}{\lambda} \frac{\varepsilon_m}{m} > 0$ in the non-pious case. These could change $g^{h'}(\lambda) < 0$ to $g^{ph'}(\lambda) > 0$.

The outcome of the standard comparative statics is again consistent with direct exploration of the derivative of $g^{ph}(\lambda) = \frac{f^{ph}\left[m^h + \frac{G_{-h}}{\lambda}, \left(\frac{1-\lambda}{\lambda}\right)G_{-h}, \frac{1}{\lambda}\right]}{\lambda n^h}$, as with its counterpart for non-pious preferences. With Cobb-Douglas utility, $g^{ph'}(1) < g^{h'}(1) = 0$. Pious agents are likely to regard the church's services and transfers to the poor as complementary. This will make $g^{ph'}(\lambda)$ more negative than $g^{h'}(\lambda)$: exactly what $g^{ph'}(1) < g^{h'}(1) = 0$ says.

Ultimately, $\text{sign}(g^{ph'}(\lambda))$ can only be resolved empirically or by explicit preferences. For example, suppose utility is again log Cobb-Douglas, now $U[x^h, G, \left(\frac{1-\lambda}{\lambda}\right)G] = \alpha_1 \log x^h + \alpha_2 \log G + \alpha_3 \log \left[1 + \left(\frac{1-\lambda}{\lambda}\right)G\right]$, $\alpha_i \in (0, 1)$, $i = 1, 2, 3$, again elasticity parameters. The additive constant, 1, in the third argument is to stop transfers to the poor satisfying an Inada condition. Substituting in the budget constraint $x^h + \frac{G}{\lambda} = m^h + \frac{G_{-h}}{\lambda}$, the arbitrary h -man chooses G to maximise $\alpha_1 \log \left(m^h + \frac{G_{-h}}{\lambda} - \frac{G}{\lambda}\right) + \alpha_2 \log G + \alpha_3 \log \left[1 + \left(\frac{1-\lambda}{\lambda}\right)G\right]$. The FOC for this generates the following quadratic equation in G :

$$\left(m^h + \frac{G_{-h}}{\lambda}\right) \lambda^2 \alpha_2 + \left[(1-\lambda) \lambda \left(m^h + \frac{G_{-h}}{\lambda}\right) (\alpha_2 + \alpha_3) - \lambda (\alpha_1 + \alpha_2)\right] G - (1-\lambda) (\alpha_1 + \alpha_2 + \alpha_3) G^2 = 0. \quad (26)$$

This can be solved for G as a function of the h -person's full income, $m^h + \frac{G_{-h}}{\lambda}$, the elasticity parameters, α_i , and expenditure shares, $(\lambda, (1-\lambda))$. Alternatively, by symmetry, using $G = \lambda n^h g^{ph}$ and $G_{-h} = \lambda (n^h - 1) g^{ph}$ in (26) and rearranging, the individual donation in a Nash equilibrium is found from the following quadratic equation defining g^{ph} implicitly in terms of λ :

$$\alpha_2 m^h + [(\alpha_2 + \alpha_3)(1-\lambda) m^h n^h - (\alpha_2 + \alpha_1 n^h)] g^{ph} - (1-\lambda) n^h (\alpha_2 + \alpha_3 + \alpha_1 n^h) g^{ph^2} = 0 \quad (27)$$

The Charity's Maximisation with Pious Preferences

In a symmetric Nash equilibrium, if only the rich donate, the charity's welfare is now

$$W^P = n^h U \left[m^h - g^h, (1 - \lambda) n^h g^h, \lambda n^h g^h \right] + n^l U \left[m^l + \frac{(1 - \lambda)}{n^l} n^h g^h, (1 - \lambda) n^h g^h, \lambda n^h g^h \right] \quad (28)$$

Let subscript T show derivatives for the T argument of $U(x, G, T)$. Simplifying with an h -type's FOC, the derivative of W^P w.r.t. λ at $\lambda = 1$ is

$$\left. \frac{\partial W^P}{\partial \lambda} \right|_{\lambda=1} = n^h g^h \left[n^h U_G^h + n^l U_G^l - n^h U_T^h - n^l U_T^l - U_x^l \right] + n^h g^{h'} \left[(n^h - 1) U_G^h + n^l U_G^l \right] \quad (29)$$

From (29) and (14) or (15), pious preferences makes poverty alleviation relatively more cost-effective. The population size effect militating against using donations to alleviate poverty in favour of public good provision no longer operates. The altruistic externality from poverty alleviation (i.e., $n^h U_T^h + n^l U_T^l$) now increases with the population just like the offsetting marginal benefit from the public good, $n^h U_G^h + n^l U_G^l$. Poverty alleviation is worthwhile if the marginal utility of the first increment of private transfers exceeds that from the marginal amount of public good when all donations are used for the latter.

I verify this insight numerically in the log-Cobb-Douglas case. By Proposition 3, $g^{ph'}(1) < 0$. So, $n^h g^{ph}(1) \left[n^h U_G^h + n^l U_G^l - n^h U_T^h - n^l U_T^l - U_x^l \right] |_{\lambda=1} \leq 0$ suffices for $\left. \frac{\partial W^P}{\partial \lambda} \right|_{\lambda=1} < 0$, by (29). Now, $g^{ph}(1) = \frac{\alpha_2 m^h}{[\alpha_2 + \alpha_1 n^h]}$ and $G^p(1) = \frac{\alpha_2 m^h n^h}{\alpha_2 + \alpha_1 n^h}$, as with non-pious preferences. Thus,

$$\begin{aligned} & n^h g^{ph}(1) \left[n^h U_G^h + n^l U_G^l - n^h U_T^h - n^l U_T^l - U_x^l \right] \quad (30) \\ &= n^h g^{ph}(1) \left[(n^h + n^l) \left(\frac{\alpha_2}{G^p} \right) - (n^h + n^l) \alpha_3 - \left(\frac{\alpha_1}{m^l} \right) \right] \\ & \left(\begin{array}{l} \geq \\ \leq \end{array} \right) 0 \text{ as } (n^h + n^l) (\alpha_2 + \alpha_1 n^h) m^l - (n^h + n^l) \alpha_3 n^h m^h m^l - \alpha_1 n^h m^h \\ &= m^l \{ (n^h + n^l) [(\alpha_2 + \alpha_1 n^h) - \alpha_3 n^h m^h] - \alpha_1 n^h \frac{m^h}{m^l} \} \left(\begin{array}{l} \geq \\ \leq \end{array} \right) 0. \end{aligned}$$

The expression in the last line of (30) is negative for plausible parameter values and, specifically, most U.S. congregation sizes. So, notably, a larger community does not make

$\left. \frac{\partial W^P}{\partial \lambda} \right|_{\lambda=1} < 0$ less likely now. It also shows a dual influence of m^h in deciding the *desirability* and *affordability* of private transfers. First, the larger is m^h relative to m^l , thus the greater is inequality and a *need* for transfers, all else equal, the more negative is $-\alpha_1 n^h \frac{m^h}{m^l} < 0$, so the more likely is $\left. \frac{\partial W^P}{\partial \lambda} \right|_{\lambda=1} < 0$. Second, as $\alpha_i \in (0, 1), i = 1 - 3$, $\text{sign}[(\alpha_2 + \alpha_1 n^h) - \alpha_3 n^h m^h]$ is determined mainly by the relative magnitudes of $\alpha_1 n^h$ and $\alpha_3 n^h m^h$, independent of n^h . All else equal, this ambiguous term is more likely negative, so $\left. \frac{\partial W^P}{\partial \lambda} \right|_{\lambda=1} < 0$ and poverty-alleviating transfers are more likely, the larger is m^h . This reflects that the larger is m^h , all else equal, the more the rich can *afford* private transfers to the poor.

Table I here

Table I exemplifies the charity optimum with pious preferences to show that it plausibly entails $1 > \lambda^p > 0$. Throughout, the congregation size is set at a realistic 180, $n^h/n^l = 5$ and each rich putative donor has five times the income of a poor putative private transfer recipient. Only preference parameters, $\alpha_i \in (0, 1), i = 1 - 3$, change between cases. In each, the charity maximises welfare W^P subject to the constraint (27) given by donors' maximising behaviour in the Nash equilibrium¹⁹. The lessons from this simple exercise are as expected:

- (i) Being able to make poverty alleviating transfers can produce a relatively large increase in total welfare. E.g., in case A, welfare increases by over 50% (from 482.22 to 730.91).
- (ii) It can result in a substantial increase in giving, to such an extent that the impact on the provision of the pure public good is negligible. For example, in case A, while just over half of the value of donations is used for private transfers to the poor, public good provision is still nearly 98% of its level without private transfers. In this case, the amount of an h person's donation that goes to public good provision is $0.4942(1.3093) = 0.64706$. This compares to a donation of 0.66225 going totally to public good provision in the case without transfers. In the other examples, the smallest ratio of pure public good provision with transfers to that without occurs in case B at nearly 90%.

¹⁹These solutions, detailed in an Online Appendix, are found numerically using Maple in Scientific Workplace. The Appendix shows that the second-order condition holds at these solutions.

(iii) It can lead to a significant increase in the poor's private consumption and living standard. In the examples, the smallest increase in their consumption occurs in C, at just over 4%; the largest occurs in A at over 16.5%.

(iv) λ is determined by the relative weight put upon the public good and private transfers in the utility function. An equal or near-equal weighting results in a λ close to 0.5 too. As the weight put on transfers (α_3) increases relative to that put on public good consumption (α_2), so the share of total giving devoted to private transfers ($1 - \lambda$) increases. So, in the simulations, as in case D, when the weight on transfers is approximately twice that on the public good, approximately twice as much of donations go to transfers as to the public good.

(v) Other things equal, the amount each rich person donates is determined by the total weighting in the utility function of transfers and the public good relative to the weighting for private consumption. The greater the weighting on private consumption, the less given.

(vi) When there is no scope for poverty alleviation, giving and welfare is independent of the exponent on private transfers in the utility function. This is seen by comparing giving and welfare in cases A and B, C and D, and E and F.

(vii) The levels of giving in the simulation can approximate those found empirically in congregations: in some examples, over 1% of aggregate income is donated, with just under half being used for poverty alleviation.

The Cobb-Douglas case shows these points clearly, but I expect such considerations to be relevant to other pious utility functions.

IV Other Preferences

Warm glow/joint-product preferences.

Would a utilitarian charity with a constituency of donors who have joint-product/warm glow preferences without piety find it cost-effective to engage in poverty alleviation? No. This is easily seen by extending the argument utilised so far.

With warm glow/joint-product preferences, $U(m^h - g^h, \lambda g^h + G_{-h}, g^h)$ is an arbitrary rich person's utility. I assume that the third argument (which yields warm glow from giving) has no Inada condition, but the second does. So, no-one is forced by their tastes to give. Let subscript w on the utility function now denote its derivative with respect to warm glow; superscript w identifies magnitudes in this case. In a Nash equilibrium in giving with a fraction $(1 - \lambda)$ of donations going to the poor, such a person has FOC

$$\begin{aligned} -U_x(m^h - g^h, \lambda g^h + G_{-h}, g^h) + \lambda U_G(m^h - g^h, \lambda g^h + G_{-h}, g^h) \\ + U_w(m^h - g^h, \lambda g^h + G_{-h}, g^h) = 0 \end{aligned} \quad (31)$$

A symmetric equilibrium with only rich donors gives a charity maximand

$$W^w = n^h U(m^h - g^h, \lambda n^h g^h, g^h) + n^l U\left(m^l + \frac{(1 - \lambda) n^h g^h}{n^l}, \lambda n^h g^h, 0\right) \quad (32)$$

Differentiating W^w and evaluating at $\lambda = 1$ using an h -person's FOC,

$$\left. \frac{\partial W^w}{\partial \lambda} \right|_{\lambda=1} = n^h g^{wh} [n^h U_G^h + n^l U_G^l - U_x^l] + n^h g^{wh'} [(n^h - 1) U_G^h + n^l U_G^l] \quad (33)$$

Equation (33) is identical in form to (14) for the charity's behaviour at $\lambda = 1$ with non-pious preferences without warm glow. Again, $\left. \frac{\partial W^w}{\partial \lambda} \right|_{\lambda=1} > 0$ is likely for even moderately-sized communities and reasonable parameters. E.g., consider the log Cobb-Douglas example with warm glow instead of utility from total transfers to the poor. Then, I can show that $g^{wh'}(1) = 0$ and, for large n^h and/or n^l , that $n^h g^{wh} [n^h U_G^h + n^l U_G^l - U_x^l] > 0$, so private transfers to the poor are not made. I summarise this in Proposition 4.

PROPOSITION 4. If consumers have identical log-Cobb-Douglas preferences with warm glow/joint-product from giving, a utilitarian charity will not make private transfers to the poor in a large community if resources can be used instead for a pure public good.

Joint-product/warm glow preferences give a generalised increased incentive for giving. This induces more giving, all else equal. But, as Proposition 4 shows, warm glow per se does not affect the cost-effectiveness of using donations for transfers to reduce poverty.

Warm glow from multiple giving

A variant on the warm-glow model is suggested, but not formally analyzed, by Reinstein (2011). He argues, from his empirical findings, that "larger givers" might have "concave multi-charity warm-glow preferences". I have examined multiple uses for a donation, albeit not multiple charities. It is intuitively plausible that some donors might wish to diversify their giving if the incremental warm glow from donating for a given purpose falls as the donation increases. So, suppose donors get different amounts of warm glow from a given gift depending upon how it is split between, say, poverty alleviation and the pastor's and church's upkeep. I make the following assumption:

(A.7). (i) Each putative donor has a concave utility function $U(x, G, \omega)$ incorporating the concave warm glow function, $\omega(\lambda g, (1-\lambda)g)$, which is increasing in each of its arguments: $\frac{\partial \omega(\lambda g, (1-\lambda)g)}{\partial (\lambda g)} \equiv \omega_1 > 0$; $\frac{\partial \omega(\lambda g, (1-\lambda)g)}{\partial ((1-\lambda)g)} \equiv \omega_2 > 0$ and $U_i(x, G, \omega) > 0$, $i = x, G, \omega$; (ii) $\omega_2(g, 0) > \omega_1(g, 0)$; (iii) $\omega_{ii} \leq 0$, $i = 1, 2$; (iv) $\omega_{ij} \geq 0$, $i, j = 1, 2$; (v) $\omega(0, 0) = 0$.

The classic Andreoni model with warm glow a linear function of aggregate giving is a special case of this formulation with $\omega_1 = \omega_2 = 1$ instead of (A.7)(ii) and $\omega_{ii} \equiv 0$, $i = 1, 2$.²⁰ I do not impose an Inada condition on ω , so someone with finite resources need not feel infinite misery from not giving. However, (A.7)(ii) means someone giving for the public good but not poverty alleviation would get increased utility if a marginal amount of their gift was shifted from public good provision to poverty alleviation.

Denote magnitudes in this case by scripts ω . Again consider a distribution such that only the rich donate. An arbitrary rich person has utility $U[m^h - g^h, \lambda(g^h + (n^h - 1)\bar{g}^h), \omega(\lambda g^h, (1-\lambda)g^h)]$.

²⁰Null (2011, note 6, p.457) suggests that the classic Andreoni case is more general than that with multi-giving warm glow. This is clearly incorrect, given my argument here.

The charity's maximand in a symmetric equilibrium is

$$W^\omega = n^h U [m^h - g^h, \lambda n^h g^h, \omega (\lambda g^h, (1 - \lambda) g^h)] + n^l U \left[m^l + \left(\frac{(1 - \lambda)}{n^l} \right) g^h, \lambda n^h g^h, \omega (0, 0) \right] \quad (34)$$

An h person's FOC in a symmetric Nash equilibrium is:

$$-U_x [m^h - g^{h\omega}, \lambda n^h g^{h\omega}, \omega (\lambda g^{h\omega}, (1 - \lambda) g^{h\omega})] + \lambda U_G^h [\cdot] + (\lambda \omega_1^h + (1 - \lambda) \omega_2^h) U_\omega^h [\cdot] = 0 \quad (35)$$

So, omitting functional arguments,

$$\begin{aligned} \frac{\partial W^\omega}{\partial \lambda} = & -n^h U_x^h g^{h\omega'} + n^h U_G^h (n^h g^{h\omega} + \lambda n^h g^{h\omega'}) + n^h U_\omega^h [\omega_1^h (g^{h\omega} + \lambda n g^{h\omega'}) + \omega_2^h ((1 - \lambda) g^{h\omega'} - g^{h\omega})] \\ & + n^l U_x^l \left\{ -\frac{n^h g^{h\omega}}{n^l} + \frac{(1 - \lambda) n^h g^{h\omega'}}{n^l} \right\} + n^l U_G^l (n^h g^{h\omega} + \lambda n^h g^{h\omega'}) \end{aligned}$$

Rearranging this and using the FOC (35) at $\lambda = 1$ yields

$$\frac{\partial W^\omega}{\partial \lambda} \Big|_{\lambda=1} = n^h g^{h\omega'} (1) [(n^h - 1) U_G^h + n^l U_G^l + U_\omega^h \omega_1^h] + n^h g^{h\omega} (1) [n^h U_G^h + n^l U_G^l + U_\omega^h (\omega_1^h - \omega_2^h) - U_x^l] \quad (36)$$

I can now show the following.

PROPOSITION 5. If agents have concave multi-giving warm glow functions $\omega (\lambda g, (1 - \lambda) g)$, then: (i) $\frac{\partial W^\omega}{\partial \lambda} \Big|_{\lambda=1}$ is ambiguously-signed; (ii) $g^{h\omega'} (1)$ is ambiguously-signed; $\frac{\partial[(\omega_1^h - \omega_2^h)/\omega^h]}{\partial g^h} \leq 0$ suffices for $g^{h\omega'} (1) < 0$ when $U[\cdot]$ is log Cobb-Douglas.

PROOF. (i) For notational ease, let $\mathbf{x}^{h\omega} (1) \equiv (m^h - g^{h\omega} (1), n^h g^{h\omega} (1), \omega (g^{h\omega} (1), 0))$ and $\mathbf{x}^{l\omega} (1) \equiv (m^l, n^h g^{h\omega} (1), \omega (0, 0))$. At $\lambda = 1$, an h person's FOC shows $\omega_1 (g^{h\omega} (1), 0) U_\omega [\mathbf{x}^{h\omega} (1)] = U_x [\mathbf{x}^{h\omega} (1)] - U_G [\mathbf{x}^{h\omega} (1)] > 0$. So, $\omega_1^h U_\omega^h$ is finite at $\lambda = 1$, as is $(\omega_1^h - \omega_2^h) U_\omega^h (< 0)$ in (36) for $\frac{\partial W^\omega}{\partial \lambda} \Big|_{\lambda=1}$. By an argument used above, for a given m^h and n^h , $U_x [\mathbf{x}^{l\omega} (1)]$ is fixed and finite. The same is true of $U_\omega [\mathbf{x}^{l\omega} (1)]$. Hence, although $-U_x [\mathbf{x}^{l\omega} (1)] + U_G [\mathbf{x}^{l\omega} (1)] + \omega_1^l U_\omega [\mathbf{x}^{l\omega} (1)] \leq 0$ holds, there exists $\hat{n}^l > 1$ so that $n^h U_G [\mathbf{x}^{h\omega} (1)] + n^l U_G [\mathbf{x}^{l\omega} (1)] + U_x [\mathbf{x}^{h\omega} (1)] (\omega_1^h - \omega_2^h) - U_x [\mathbf{x}^{l\omega} (1)] > 0$, thus the $g^{h\omega} (1)$ term in $\frac{\partial W^\omega}{\partial \lambda} \Big|_{\lambda=1}$ is positive, for $n^l \geq \hat{n}^l$. So, $\frac{\partial W^\omega}{\partial \lambda} \Big|_{\lambda=1}$ is ambiguous and $g^{h\omega'} < 0$ is necessary but not sufficient for $\frac{\partial W^\omega}{\partial \lambda} \Big|_{\lambda=1} < 0$

when n^h and/or n^l is "large". ■

(ii) To sign the term in $g^{h\omega'}$, again consider a neighbourhood of $\lambda = 1$ where $g^{h\omega}(1) > 0 = g^{l\omega}(1)$. Then, differentiating through an h person's FOC (35), rearranging and evaluating at $\lambda = 1$ gives

$$g^{h\omega'}(1) = - \frac{\{g^{h\omega}(1) [n^h U_{GG}^h + \omega_1^h (\omega_1^h - \omega_2^h) U_{\omega\omega}^h + (\omega_{11}^h - \omega_{12}^h) U_{\omega}^h] + (\omega_1^h - \omega_2^h) U_{\omega}^h + U_G^h\}}{D} \quad (37)$$

Now, $D \equiv [U_{xx}^h + \omega_1^{h2} U_{\omega\omega}^h + \omega_{11}^h U_{\omega}^h + n^h U_{GG}^h] < 0$ by concavity. However, the sign of the numerator of $g^{h\omega'}(1)$ is a priori ambiguous, as is $g^{h\omega'}(1)$ therefore. For tractability and comparability, consider a log-Cobb-Douglas $U[\cdot]$. Then $n^h g^{h\omega}(1) U_{GG}^h + U_G^h = 0$. Thus,

$$g^{h\omega'}(1) = - \frac{\{g^{h\omega}(1) [\omega_1^h (\omega_1^h - \omega_2^h) U_{\omega\omega}^h + (\omega_{11}^h - \omega_{12}^h) U_{\omega}^h] + (\omega_1^h - \omega_2^h) U_{\omega}^h\}}{D}.$$

Likewise, $U_{\omega\omega}^h = -\frac{U_{\omega}^h}{\omega^h}$ as $U[\cdot]$ is log-Cobb-Douglas. So,

$$\begin{aligned} \omega_1^h (\omega_1^h - \omega_2^h) U_{\omega\omega}^h + (\omega_{11}^h - \omega_{12}^h) U_{\omega}^h &= -\omega_1^h (\omega_1^h - \omega_2^h) \frac{U_{\omega}^h}{\omega^h} + (\omega_{11}^h - \omega_{12}^h) U_{\omega}^h \\ &= \left(\frac{U_{\omega}^h}{\omega^h}\right) [(\omega_{11}^h - \omega_{12}^h) \omega^h - \omega_1^h (\omega_1^h - \omega_2^h)] = \omega^h U_{\omega}^h \frac{\partial[(\omega_1^h - \omega_2^h)(\omega^h)^{-1}]}{\partial g^h} \leq 0, \end{aligned}$$

all these magnitudes being evaluated at $\omega(g^{h\omega}(1), 0)$. Thus, as $(\omega_1^h - \omega_2^h) < 0$ at $\lambda = 1$ from (A.7), $\frac{\partial[(\omega_1^h - \omega_2^h)(\omega^h)^{-1}]}{\partial g^h} \leq 0$ suffices for $g^{h\omega'}(1) < 0$ in the log-Cobb-Douglas case. ■

Now, $\omega_1(g^{h\omega}(1), 0) - \omega_2(g^{h\omega}(1), 0)$ is the marginal incentive to give for the public good instead of poverty alleviation if no alleviation occurs; $\frac{[\omega_1(g^{h\omega}(1), 0) - \omega_2(g^{h\omega}(1), 0)]}{\omega(g^{h\omega}(1), 0)}$ is that incentive relative to warm glow then. So, intuitively, it suffices for $g^{h\omega'}(1) < 0$ that warm glow from public good giving declines relative to that from transfers the larger is public good giving and warm glow from it already. If a donor's warm glow is linear in total giving irrespective of its uses, with log-Cobb-Douglas utility, $g^{h\omega'}(1) = 0$ and, for large n^h and/or n^l , $\frac{\partial W^{\omega}}{\partial \lambda}|_{\lambda=1} > 0$. With concave multi-giving warm glow, $g^{\omega'}(1) < 0$ might hold even if $\frac{\partial W^{\omega}}{\partial \lambda}|_{\lambda=1} \geq 0$ does. I.e., a charity might increase giving if it uses part of donations for transfers, but this need not increase welfare. Anyway, multi-giving concave warm glow makes outcomes for poverty reduction and welfare less predictable than if warm glow just depends on total giving.

Brown (1997) preferences.

In Brown's model, rich donors get an altruistic externality from giving that depends on the poor's per capita income.²¹ Brown does not consider the poor's welfare explicitly. But, suppose we do when resources used for private transfers to them can instead be used for a pure public good. I can show that, even if everyone (including the poor) gets an altruistic externality from giving that alters the poor's per capita income, there is unlikely to be enough incentive for poverty alleviation in typically-sized giving communities like congregations.

Indexing this case by "B", the charity now maximises welfare given by

$$\begin{aligned} W^B = & n^h U \left(m^h - g^h, \lambda n^h g^h, m^l + \frac{(1-\lambda) n^h g^h}{n^l} \right) \\ & + n^l U \left(m^l + \frac{(1-\lambda) n^h g^h}{n^l}, \lambda n^h g^h, m^l + \frac{(1-\lambda) n^h g^h}{n^l} \right). \end{aligned} \quad (38)$$

Letting subscripts s denote derivatives for the poor's per capita income,

$$\begin{aligned} \left. \frac{\partial W^B}{\partial \lambda} \right|_{\lambda=1} = & n^h g^{Bh} \left[n^h U_G^h + n^l U_G^l - \left(\frac{n^h}{n^l} \right) U_s^h - U_s^l - U_x^l \right] \\ & + n^h g^{Bh'} \left[(n^h - 1) U_G^h + n^l U_G^l \right] \end{aligned} \quad (39)$$

The poor's living standard, their per capita income, is now an externality for all community members. Still, this is insufficient to ensure private transfers to the poor are cost-effective in large communities. The reason is evident from (39). Each donated dollar used for private transfers is split equally among the poor, raising their per capita income by $(n^l)^{-1}$. This yields total altruistic externality of $n^l U_s^l (n^l)^{-1} = U_s^l$ for the poor and $n^h U_s^h (n^l)^{-1}$ for the rich, compared with $n^l U_s^l$ and $n^h U_s^h$, respectively, with pious preferences. Conversely, using the same dollar instead for pure public good provision gives benefits that rise in proportion to the sizes of the rich and poor communities, as with pious preferences. This make

²¹This feature links Brown's model to Atkinson (2009).

poverty alleviation less likely with Brown preferences than pious preferences. This can be again illustrated by log-Cobb-Douglas preferences. It holds although $g^{Bh'}(1) < 0$ is true in the log-Cobb-Douglas case and can be shown to hold generally under the same condition ($-G^B \frac{U_{GG}^h}{U_G^h} \geq 1$) as with pious preferences. I summarise this in Proposition 6:

PROPOSITION 6. If utility is log-Cobb-Douglas and increases with the poor's per capita consumption, then, if funds can be used instead for a pure public good, a utilitarian charity is less likely to make private transfers to the poor if consumers have Brown preferences than it would if they have pious preferences.

V Conclusion

Becker, in his "A Theory of Social Interactions" (1974), quotes *The Random House Dictionary of the English Language's* (1967, unabridged) definition of charity: "the benevolent feeling, especially towards those in need or in disfavour." He includes the recipients' well-being in the donors' utility functions to study explicitly the trade-off between a recipient's income and a charitable person's giving. However, explicit analyses of charitable transfers to the needy are conspicuously absent in much of the theoretical charity literature in the subsequent 40 years. I have shown a difficulty of incorporating such transfers in models based on utility functions common in that literature. This difficulty arises as a large-population effect makes poverty-alleviating private transfers cost-ineffective if the funds could instead be used for a public good. I also show that this difficulty is overcome if we posit preferences close to the spirit of both Becker's and Hochman and Rodgers' original conceptions. Such preferences internalise the amount the community gives to the poor and so also reflect the religious precepts that one is one's brothers' keeper and should love thy neighbours like yourself. Hence I label them "pious preferences" for short.

Several authors (e.g., Feldstein, 1975; List, 2011; Andreoni and Payne, 2013) suggest, from empirical findings, that religious giving is different from other forms. For example, it is

less susceptible to vagaries of the economic cycle (List, 2011), is less price elastic (Feldstein, 1975, and followers) and the amount given depends on the devoutness of the giver (The Barna Group, 2009; McKittrick et al, 2013). Such findings suggest that the motivation for religious giving is not narrowly economic. I have captured this aspect of religious giving by modeling agents who empathize with their poorer brethren by having utility that responds positively to the total amount given to the latter. As "eleven o'clock Sunday morning continues to be the most segregated hour in America" (Hartford Institute for Religion Research, 2015), I believe that this empathy is also important for reasons I have not stressed: religious giving allows donors to improve the well-being only of people that they identify with. So, it is bound up with the effect that Alesina *et al* (1999) identify, wherein giving within essentially homogeneous communities substitutes for and undermines more widely distributed state provision. This is an issue to be pursued in further research. The same is true of the exploration of any complementarity or substitutability in the relationship between joint-product/warm-glow giving and pious preferences.

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Table 1. Illustrative Simulations of Giving with Pious Preferences

$m^h = 100; m^l = 20; n^h = 150; n^l = 30$						
Case	Parameters	Giving				
		With Transfers			Without transfers	
		Giving	Welfare	l 's % consumption increase	Giving (g)	Welfare
A	$\alpha_1 = 0.3; \alpha_2 = 0.3;$	$g = 1.3093$	730.91	16.56	0.662 25	482.22
	$\alpha_3 = 0.3$	$\lambda = 0.4942$				
B	$\alpha_1 = 0.3; \alpha_2 = 0.3;$	$g = 1.2016$	680.93	15.17	0.662 25	482.22
	$\alpha_3 = 0.25$	$\lambda = 0.49504$				
C	$\alpha_1 = 0.4; \alpha_2 = 0.1;$	$g = 0.32579$	428.31	4.12	0.166 39	370.07
	$\alpha_3 = 0.1$	$\lambda = 0.49363$				
D	$\alpha_1 = 0.4; \alpha_2 = 0.1;$	$g = 0.49098$	440.82	8.25	0.166 39	370.07
	$\alpha_3 = 0.2$	$\lambda = 0.32778$				
E	$\alpha_1 = 0.45; \alpha_2 = 0.225;$	$g = 0.65574$	668.26	8.30	0.33232	509.37
	$\alpha_3 = 0.225$	$\lambda = 0.49374$				
F	$\alpha_1 = 0.45; \alpha_2 = 0.225;$	$g = 0.63390$	655.07	7.81	0.33232	509.37
	$\alpha_3 = 0.21$	$\lambda = 0.5073$				

Online Appendix to "Faith, Hope and Christian Charity: How Religion Explains Giving when Warm Glow and Impure Altruism Do Not" by Clive D. Fraser

For the numerical solutions reported in the paper, the Charity solves the following constrained optimisation problem:

$$\begin{aligned}
 \text{Max}_{g, \lambda} M(g, \lambda) &\equiv n^h [\alpha_1 \ln(m^h - g^h) + \alpha_2 \ln \lambda n^h g^h + \alpha_3 \ln(1 + (\frac{1-\lambda}{\lambda}) \lambda n^h g^h)] + \\
 &n^l [\alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + \alpha_2 \ln \lambda n^h g^h + \alpha_3 \ln(1 + (\frac{1-\lambda}{\lambda}) \lambda n^h g^h)] \\
 &= n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) +
 \end{aligned}$$

$$(n^h + n^l) \alpha_2 \ln \lambda n^h g^h + (n^h + n^l) \alpha_3 \ln (1 + (1 - \lambda) n^h g^h) \quad (40)$$

subject to the constraint from the households' maximisation if only the h -type gives,

$$C(g, \lambda) = \alpha_2 m^h + [(\alpha_2 + \alpha_3)(1 - \lambda) m^h n^h - (\alpha_2 + \alpha_1 n^h)] g^h - (1 - \lambda) n^h (\alpha_2 + \alpha_3 + \alpha_1 n^h) g^{h^2} = 0 \quad (41)$$

Forming the Lagrangian $L = M(g, \lambda) - \mu C(g, \lambda)$, where μ is a Lagrange multiplier, the FOC for this maximisation are the constraint (2) and

$$\nabla M(g, \lambda) = \mu \nabla C(g, \lambda) \quad (42)$$

I.e., omitting the script in g^h , the FOC are $\nabla L = \nabla M(g, \lambda) - \mu \nabla C(g, \lambda) = 0$; $C(g, \lambda) = 0$, where

$$\nabla M(g, \lambda) = \begin{pmatrix} \frac{\partial M}{\partial g} \\ \frac{\partial M}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} -\frac{n^h \alpha_1}{(m^h - g)} + \frac{(1 - \lambda) \alpha_1 n^h n^l}{(m^l n^l + (1 - \lambda) n^h g)} + \frac{(n^h + n^l) \alpha_2}{g} + \frac{\alpha_3 (n^h + n^l) (1 - \lambda) n^h}{(1 + (1 - \lambda) n^h g)} \\ -\frac{\alpha_1 n^h n^l g}{(m^l n^l + (1 - \lambda) n^h g)} + \frac{(n^h + n^l) \alpha_2}{\lambda} - \frac{\alpha_3 (n^h + n^l) n^h g}{(1 + (1 - \lambda) n^h g)} \end{pmatrix}$$

and

$$\mu \nabla C(g, \lambda) = \mu \begin{pmatrix} \frac{\partial C}{\partial g} \\ \frac{\partial C}{\partial \lambda} \end{pmatrix} = \begin{pmatrix} \mu ((1 - \lambda) ((\alpha_2 + \alpha_3) m^h n^h - 2n^h (\alpha_2 + \alpha_3 + \alpha_1 n^h) g) - (\alpha_2 + \alpha_1 n^h)) \\ \mu (-(\alpha_2 + \alpha_3) m^h n^h g + n^h (\alpha_2 + \alpha_3 + \alpha_1 n^h) g^2) \end{pmatrix}$$

For the second-order condition (SOC), from Murata (1977) Theorem 36 Corollary (p59 and pp.267-8), the Hessian of the Lagrangian $L = M(g, \lambda) - \mu C(g, \lambda)$ must be negative

definite for all variations δ satisfying $\nabla C' \delta = 0$. This is so *iff*
$$\begin{vmatrix} 0 & \psi_1 & \psi_2 \\ \psi_1 & \psi_{11} & \psi_{12} \\ \psi_2 & \psi_{12} & \psi_{22} \end{vmatrix} > 0,$$

where $\psi_1 = \partial C(g, \lambda) / \partial g$;

$\psi_2 = \partial C(g, \lambda) / \partial \lambda$;

$\psi_{11} = \partial^2 L / \partial g^2 = \partial^2 M(g, \lambda) / \partial g^2 - \mu \partial^2 C(g, \lambda) / \partial g^2$;

$\psi_{12} = \partial^2 L / \partial g \partial \lambda = \partial^2 M(g, \lambda) / \partial g \partial \lambda - \mu \partial^2 C(g, \lambda) / \partial g \partial \lambda$;

$$\psi_{22} = \partial^2 L / \partial \lambda^2 = \partial^2 M(g, \lambda) / \partial \lambda^2 - \mu \partial^2 C(g, \lambda) / \partial \lambda^2.$$

I assume $m^h = 100; m^l = 20; n^h = 150, n^l = 30$ in the numerical examples. So,

$$\psi_1 = \partial C(g, \lambda) / \partial g = (\alpha_2 + \alpha_3)(1 - \lambda)15000 - (\alpha_2 + \alpha_1 150) - (1 - \lambda)300(\alpha_2 + \alpha_3 + 150\alpha_1)$$

$$\psi_2 = \partial C(g, \lambda) / \partial \lambda = -(\alpha_2 + \alpha_3)15000g + 150(\alpha_2 + \alpha_3 + 150\alpha_1)g^2$$

$$\psi_{11} = \partial^2 L / \partial g^2 = -\frac{150\alpha_1}{(100-g)^2} - \frac{30\alpha_1(1-\lambda)^2 22500}{(600+(1-\lambda)150g)^2} - \frac{\alpha_2 180}{g^2} - \frac{180\alpha_3(1-\lambda)^2 22500}{(1+(1-\lambda)150g)^2} - \mu 300(1-\lambda)(\alpha_2 + \alpha_3 + 150\alpha_1)$$

$$\psi_{12} = \partial^2 L / \partial g \partial \lambda = -\frac{4500\alpha_1}{(600+(1-\lambda)150g)} + \frac{22500\alpha_1(1-\lambda)30g}{(600+(1-\lambda)150g)^2} - \frac{180\alpha_3 150}{(1+(1-\lambda)150g)} + \frac{180\alpha_3(1-\lambda)g 22500}{(1+(1-\lambda)150g)^2} -$$

$$\mu(150(\alpha_2 + \alpha_3 + 150\alpha_1)2g - (\alpha_2 + \alpha_3)15000)$$

$$\psi_{22} = \partial^2 L / \partial \lambda^2 = \partial^2 M(g, \lambda) / \partial \lambda^2 - \mu \partial^2 C(g, \lambda) / \partial \lambda^2 = \frac{22500\alpha_1 30g^2}{(600+(1-\lambda)150g)^2} - \frac{\alpha_2 180}{\lambda^2} +$$

$$\frac{22500\alpha_3 180g^2}{(1+(1-\lambda)150g)^2}$$

Example A

$\alpha_1 = 0.3; \alpha_2 = 0.3; \alpha_3 = 0.3; m^h = 100; m^l = 20; n^h = 150, n^l = 30$. The charity's maximand is

$$M(g, \lambda) = n^h [\alpha_1 \ln(m^h - g^h) + \alpha_2 \ln(\lambda n^h g^h) + \alpha_3 \ln(1 + (1 - \lambda)n^h g^h)]$$

$$+ n^l [\alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l})n^h g^h) + \alpha_2 \ln(\lambda n^h g^h) + \alpha_3 \ln(1 + (1 - \lambda)n^h g^h)]$$

$$= n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l})n^h g^h) + (n^h + n^l) \alpha_2 \ln(\lambda n^h g^h)$$

$$+ (n^h + n^l) \alpha_3 \ln(1 + (1 - \lambda)n^h g^h)$$

$$= 45 \ln(100 - g) + 54 \ln(\lambda 150g) + 54 \ln(1 + (1 - \lambda)150g) + 9 \ln(20 + (1 - \lambda)5g)$$

$$\frac{\partial M(g, \lambda)}{\partial g} = -\frac{n^h \alpha_1}{(m^h - g)} + \frac{(n^h + n^l) \alpha_2}{g} + \frac{(1-\lambda) \alpha_1 n^h n^l}{(m^l n^l + (1-\lambda)n^h g)} + \frac{\alpha_3 (n^h + n^l)(1-\lambda)n^h}{(1+(1-\lambda)n^h g)}$$

$$= -\frac{45}{100-g} + \frac{54}{g} + 54 \frac{150(1-\lambda)}{1+150(1-\lambda)g} + \frac{45(1-\lambda)}{20+5(1-\lambda)g}$$

$$\frac{\partial M(g, \lambda)}{\partial \lambda} = -\frac{\alpha_1 n^h n^l g}{(m^l n^l + (1-\lambda)n^h g)} + \frac{(n^h + n^l) \alpha_2}{\lambda} - \frac{\alpha_3 (n^h + n^l) n^h g}{(1+(1-\lambda)n^h g)}$$

$$= \frac{54}{\lambda} - 54 \frac{150g}{1+150(1-\lambda)g} - \frac{45g}{20+5(1-\lambda)g}$$

$$C(g, \lambda) = 30 + 9000g - 9000g\lambda - 45.3g - 6840g^2 + 6840g^2\lambda$$

$$= 30 + (1 - \lambda)(9000g - 6840g^2) - 45.3g$$

$$\frac{\partial C(g, \lambda)}{\partial g} = 9000 - 9000\lambda - 45.3 - 13680g + 13680g\lambda$$

$$= (1 - \lambda)(9000 - 13680g) - 45.3$$

$$\frac{\partial C(g, \lambda)}{\partial \lambda} = -9000g + 6840g^2$$

The FOC then yield the following simultaneous equation system to be solved:

$$-\frac{45}{100-g} + \frac{54}{g} + 54 \frac{150(1-\lambda)}{1+150(1-\lambda)g} + \frac{45(1-\lambda)}{20+5(1-\lambda)g} = \mu((9000-13680g)(1-\lambda) - 45.3) \quad (43)$$

$$\frac{54}{\lambda} - 54 \frac{150g}{1+150(1-\lambda)g} - \frac{45g}{20+5(1-\lambda)g} = \mu(6840g^2 - 9000g) \quad (44)$$

$$30 + (1-\lambda)(9000g - 6840g^2) - 45.3g = 0 \quad (45)$$

Using Maple, the solution is:

$$\left\{ \begin{array}{l} [g = 1.0184 - 8.4309i, \lambda = 1.0 + 7.7937 \times 10^{-4}i, \mu = 1.8053 - 0.15381i], \\ [g = 1.0184 + 8.4309i, \lambda = 1.0 - 7.7937 \times 10^{-4}i, \mu = 1.8053 + 0.15381i], \\ [g = 1.3093, \lambda = 0.49418, \mu = -1.8141 \times 10^{-2}], \\ [g = 1.3170, \lambda = 3.7713, \mu = 4.1802 \times 10^{-4}] \end{array} \right\}$$

The Second-order Condition for Example A

With $g = 1.3093; \lambda = 0.4942 \quad \mu = -1.8141 \times 10^{-2}$ being the only sensible real solution,

$$\begin{aligned} \psi_1 &= \partial C(g, \lambda) / \partial g = (\alpha_2 + \alpha_3)(1-\lambda)15000 - (\alpha_2 + \alpha_1 150) - (1-\lambda)300(\alpha_2 + \alpha_3 + 150\alpha_1)1.3093 \\ &= 0.6(1-0.4942)15000 - (0.3 + 45) - (1-0.4942)300(0.6 + 45)1.3093 = -4552.6 \\ \psi_2 &= \partial C(g, \lambda) / \partial \lambda = -(\alpha_2 + \alpha_3)15000g + 150(\alpha_2 + \alpha_3 + 150\alpha_1)g^2 = -(0.6)15000(1.3093) + \\ &150(0.6 + 45)(1.3093^2) = -58.117 \end{aligned}$$

$$\begin{aligned} \psi_{11} &= \partial^2 L / \partial g^2 = -\frac{150\alpha_1}{(100-g)^2} - \frac{30\alpha_1(1-\lambda)^2 22500}{(600+(1-\lambda)150(g))^2} - \frac{\alpha_2 180}{g^2} - \frac{180\alpha_3(1-\lambda)^2 22500}{(1+(1-\lambda)150g)^2} - \mu 300(1-\lambda)(\alpha_2 + \alpha_3 + 150\alpha_1) \\ &= -\frac{45}{(100-1.3093)^2} - \frac{9(1-0.4942)^2 22500}{(600+(1-0.4942)150(1.3093))^2} - \frac{54}{1.3093^2} - \frac{54(1-0.4942)^2 22500}{(1+(1-0.4942)150(1.3093))^2} \\ &+ (1.8141 \times 10^{-2})300(1-0.4942)(0.6 + 45) = 63.037 \\ \psi_{12} &= \partial^2 L / \partial g \partial \lambda = -\frac{4500\alpha_1}{(600+(1-\lambda)150g)} + \frac{22500\alpha_1(1-\lambda)30g}{(600+(1-\lambda)150g)^2} - \frac{180\alpha_3 150}{(1+(1-\lambda)150g)} + \frac{180\alpha_3(1-\lambda)g 22500}{(1+(1-\lambda)150g)^2} - \\ &\mu(150(\alpha_2 + \alpha_3 + 150\alpha_1)2g - (\alpha_2 + \alpha_3)15000) \end{aligned}$$

$$= -\frac{1350}{(600+(1-0.4942)150(1.3093))} + \frac{6750(1-0.4942)30(1.3093)}{(600+(1-0.4942)150(1.3093))^2} - \frac{(54)150}{(1+(1-0.4942)150(1.3093))} + \frac{54(1-0.4942)(1.3093)22500}{(1+(1-0.4942)150(1.3093))^2} +$$

$$(1.8141 \times 10^{-2}) (150 (0.6 + 45) 2 (1.3093) - (0.6) 15000) = 159.20$$

$$\psi_{22} = \partial^2 L / \partial \lambda^2 = \frac{22500\alpha_1 30g^2}{(600+(1-\lambda)150g)^2} - \frac{\alpha_2 180}{\lambda^2} + \frac{22500\alpha_3 180g^2}{(1+(1-\lambda)150g)^2}$$

$$= \frac{(6750)30(1.3093)^2}{(600+(1-0.4942)150(1.3093))^2} - \frac{54}{(0.4942)^2} + \frac{(6750)180(1.3093)^2}{(1+(1-0.4942)150(1.3093))^2} = -13.502$$

So, the determinant of the bordered Hessian matrix is
$$\begin{vmatrix} 0 & -4552.6 & -58.117 \\ -4552.6 & 63.037 & 159.20 \\ -58.117 & 159.20 & -13.502 \end{vmatrix} =$$

$$3.6388 \times 10^8 > 0$$

Hence, this solution identifies a maximum point.

Welfare

With $g = 1.3093, \lambda = 0.49418$, Welfare is given by $45 \ln(100 - g) + 54 \ln(\lambda 150g) +$

$$54 \ln(1 + (1 - \lambda) 150g) + 9 \ln(20 + (1 - \lambda) 5g)$$

$$= 45 \ln(100 - 1.3093) + 54 \ln((0.49418)150(1.3093)) + 54 \ln(1 + (1 - 0.49418) 150(1.3093)) +$$

$$9 \ln(20 + (1 - 0.49418) 5(1.3093)) = 730.91.$$

With no poverty alleviation, $\lambda = 1$ and $g = \frac{\alpha_2 m^h}{\alpha_2 + \alpha_1 n^h}$, so $g = 0.66225$ now. Welfare is given by

$$45 \ln(100 - g) + 54 \ln(\lambda 150g) + 54 \ln(1 + (1 - \lambda) 150g) + 9 \ln(20 + (1 - \lambda) 5g)$$

$$= 45 \ln(100 - 0.66225) + 54 \ln(150(0.66225)) + 54 \ln(1) + 9 \ln(20) = 482.22 < 730.91$$

Transfers

With $g = 1.3093; \lambda = 0.4942$, the transfer to each poor person $= \frac{150(1-0.4942)1.3093}{30} =$

3.3112. This is significant as it is $\frac{3.3112}{20}(100\%) = 16.556\%$ of the poor person's original income.

Example B:

$\alpha_1 = 0.3; \alpha_2 = 0.3; \alpha_3 = 0.25 \quad m^h = 100; m^l = 20 \quad n^h = 150; n^l = 30$ The FOC system to be solved is

$$-\frac{45}{(100-g)} + \frac{(1-\lambda)9}{(4+(1-\lambda)g)} + \frac{54}{g} + \frac{45(1-\lambda)150}{(1+(1-\lambda)150g)} = \mu((1-\lambda)(8250-13665g)-45.3) \quad (46)$$

$$-\frac{9g}{(4+(1-\lambda)g)} + \frac{54}{\lambda} - \frac{(54)150g}{(1+(1-\lambda)150g)} = \mu(6832.5g^2-8250g) \quad (47)$$

$$30+(1-\lambda)(8250g-6832.5g^2)-45.3g=0 \quad (48)$$

The solution is:

$$\left\{ \begin{array}{l} [g = -6.0678 \times 10^{-4}, \lambda = -4.9953, \mu = 1.9013 \times 10^{-3}], \\ [g = 1.2016, \lambda = 0.49504, \mu = -1.9785 \times 10^{-2}], \\ [g = 0.93475 - 1.9147i, \lambda = 1.0005 + 3.0661 \times 10^{-3}i, \mu = 2.1778 - 4.1699 \times 10^{-2}i], \\ [g = 0.93475 + 1.9147i, \lambda = 1.0005 - 3.0661 \times 10^{-3}i, \mu = 2.1778 + 4.1699 \times 10^{-2}i], \\ [g = 1.2085, \lambda = 4.0227, \mu = 1.2812 \times 10^{-4}] \end{array} \right\}$$

Hence the only sensible real solution is $[g = 1.2016, \lambda = 0.49504, \mu = -1.9785 \times 10^{-2}]$.

Checking the Second-Order Condition for Example B

$$\begin{aligned} \psi_1 &= \partial C(g, \lambda) / \partial g = (\alpha_2 + \alpha_3)(1-\lambda)15000 - (\alpha_2 + \alpha_1 150) - (1-\lambda)300(\alpha_2 + \alpha_3 + 150\alpha_1)g \\ &= (0.55)(1-0.49504)15000 - (0.3+45) - (1-0.49504)300(0.55+45)(1.2016) = \\ &-4170.8 \end{aligned}$$

$$\psi_2 = \partial C(g, \lambda) / \partial \lambda = -(0.55)15000(1.2016) + 150(0.55+45)(1.2016)^2 = -48.146$$

$$\begin{aligned} \psi_{11} &= \partial^2 L / \partial g^2 = -\frac{45}{(100-(1.2016))^2} - \frac{9(1-0.49504)^2 22500}{(600+(1-0.49504)150(1.2016))^2} - \frac{54}{(1.2016)^2} - \frac{54(1-0.49504)^2 22500}{(1+(1-0.49504)150(1.2016))^2} + \\ &((.019785))300(1-0.49504)(0.55+45) = 62.417 \end{aligned}$$

$$\begin{aligned} \psi_{12} &= \partial^2 L / \partial g \partial \lambda = -\frac{1350}{(600+(1-0.49504)150(1.2016))} + \frac{6750(1-0.49504)30(1.2016)}{(600+(1-0.49504)150(1.2016))^2} - \frac{(54)150}{(1+(1-0.49504)150(1.2016))} + \\ &\frac{54(1-0.49504)(1.2016)22500}{(1+(1-0.49504)150(1.2016))^2} + (.019785)(300(45.55)(1.2016) - (0.55)15000) = 158.99 \end{aligned}$$

$$\begin{aligned} \psi_{22} &= \partial^2 L / \partial \lambda^2 = \partial^2 M(g, \lambda) / \partial \lambda^2 - \mu \partial^2 C(g, \lambda) / \partial \lambda^2 = \frac{(6750)30(1.2016)^2}{(600+(1-0.49504)150(1.2016))^2} - \\ &\frac{54}{0.49504^2} + \frac{(6750)180(1.2016)^2}{(1+(1-0.49504)150(1.2016))^2} = -12.538 \end{aligned}$$

$$\begin{vmatrix} 0 & -4170.8 & -48.146 \\ -4170.8 & 62.417 & 158.99 \\ -48.146 & 158.99 & -12.538 \end{vmatrix} = 2.8181 \times 10^8 > 0$$
 So, a maximum point has been identified.

Welfare

Without poverty alleviation, $g = \frac{\alpha_2 m^h}{\alpha_2 + \alpha_1 n^h} = \frac{30}{0.3+45} = 0.66225$.

Now, welfare $= n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + (n^h + n^l) \alpha_2 \ln \lambda n^h g^h + (n^h + n^l) \alpha_3 \ln(1 + (1 - \lambda) n^h g^h)$

$$= 45 \ln(100 - 0.66225) + 9 \ln(20) + 54 \ln(150(0.66225)) + (180) 0.25 \ln(1) = 482.22$$

With poverty alleviation, welfare $= n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + (n^h + n^l) \alpha_2 \ln \lambda n^h g^h + (n^h + n^l) \alpha_3 \ln(1 + (1 - \lambda) n^h g^h)$

$$= 45 \ln(100 - 1.2016) + 9 \ln(20 + (1 - 0.495) 5(1.2016)) + 54 \ln((0.495) 150(1.2016)) + 45 \ln(1 + (1 - 0.495) 150(1.2016)) = 680.93$$

This is obviously a significant improvement on the case without transfers.

Example C:

$\alpha_1 = 0.4; \alpha_2 = 0.1; \alpha_3 = 0.1$ $m^h = 100; m^l = 20$ $n^h = 150; n^l = 30$ After simplification of some terms, the system of FOC is

$$\frac{(1-\lambda)12}{4+(1-\lambda)g} - \frac{60}{100-g} + \frac{18}{g} + \frac{(1-\lambda)2700}{1+(1-\lambda)150g} = \mu((1-\lambda)(3000 - 18060g) - 60.1) \quad (49)$$

$$-\frac{12g}{(4+(1-\lambda)g)} + \frac{18}{\lambda} - \frac{2700g}{(1+(1-\lambda)150g)} = \mu(-3000g + 9030g^2) \quad (50)$$

$$10 + (1-\lambda)(3000g - 9030g^2) - 60.1g = 0 \quad (51)$$

The solution of this equation system is:

$$\left\{ \begin{array}{l} [g = 0.332\,60, \lambda = 9.904\,2, \mu = 1.822\,3 \times 10^{-4}], \\ [g = 0.325\,79, \lambda = 0.493\,63, \mu = -0.071\,83], \\ [g = 0.205\,03 - 2.788\,0i, \lambda = 1.0 + 2.378\,3 \times 10^{-3}i, \mu = 1.802 - 5.046\,3 \times 10^{-2}i], \\ [g = 0.205\,03 + 2.788\,0i, \lambda = 1.0 - 2.378\,3 \times 10^{-3}i, \mu = 1.802 + 5.046\,3 \times 10^{-2}i] \end{array} \right\}$$

Hence, $g = 0.325\,79, \lambda = 0.493\,63, \mu = -0.071\,83$ is the only sensible real solution.

The Second-Order Condition for Example C

To check for the SOC with $\alpha_1 = 0.4; \alpha_2 = 0.1; \alpha_3 = 0.1$,

$$\begin{aligned} \psi_1 &= \partial C(g, \lambda) / \partial g = (\alpha_2 + \alpha_3)(1 - \lambda)15000 - (\alpha_2 + \alpha_1 150) - (1 - \lambda)300(\alpha_2 + \alpha_3 + 150\alpha_1)g \\ &= (0.2)(1 - 0.493\,63)15000 - (0.1 + 60) - (1 - 0.493\,63)300(0.2 + 60)(0.325\,79) \\ &= -1520.4 \end{aligned}$$

$$\begin{aligned} \psi_2 &= \partial C(g, \lambda) / \partial \lambda = -(\alpha_2 + \alpha_3)15000g + 150(\alpha_2 + \alpha_3 + 150\alpha_1)g^2 = -(0.2)15000(0.325\,79) + \\ &150(0.2 + 60)(0.325\,79)^2 = -18.934 \end{aligned}$$

$$\begin{aligned} \psi_{11} &= \partial^2 L / \partial g^2 = -\frac{150\alpha_1}{(100-g)^2} - \frac{30\alpha_1(1-\lambda)^2 22500}{(600+(1-\lambda)150(g))^2} - \frac{\alpha_2 180}{g^2} - \frac{180\alpha_3(1-\lambda)^2 22500}{(1+(1-\lambda)150g)^2} - \mu 300(1-\lambda)(\alpha_2 + \alpha_3 + 150\alpha_1) \\ &= -\frac{60}{(100-(0.325\,79))^2} - \frac{12(1-0.493\,63)^2 22500}{(600+(1-0.493\,63)150(0.325\,79))^2} - \frac{18}{(0.325\,79)^2} - \frac{18(1-0.493\,63)^2 22500}{(1+(1-0.493\,63)150(0.325\,79))^2} - \\ &(-0.071\,83)300(1-0.493\,63)(0.2+60) = 330.45 \end{aligned}$$

$$\begin{aligned} \psi_{12} &= \partial^2 L / \partial g \partial \lambda = -\frac{4500\alpha_1}{(600+(1-\lambda)150g)} + \frac{22500\alpha_1(1-\lambda)30g}{(600+(1-\lambda)150g)^2} - \frac{180\alpha_3 150}{(1+(1-\lambda)150g)} + \frac{180\alpha_3(1-\lambda)g 22500}{(1+(1-\lambda)150g)^2} - \\ &\mu(150(\alpha_2 + \alpha_3 + 150\alpha_1)2g - (\alpha_2 + \alpha_3)15000) \\ &= -\frac{1800}{(600+(1-0.493\,63)150(0.325\,79))} + \frac{2250(1-0.493\,63)30(0.325\,79)}{(600+(1-0.493\,63)150(0.325\,79))^2} - \frac{(18)150}{(1+(1-0.493\,63)150(0.325\,79))} + \\ &\frac{18(1-0.493\,63)(0.325\,79)22500}{(1+(1-0.493\,63)150(0.325\,79))^2} - (-0.071\,83)(150(0.2+60)2(0.325\,79) - (0.2)15000) = 200. \end{aligned}$$

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$$\begin{aligned} \psi_{22} &= \partial^2 L / \partial \lambda^2 = \partial^2 M(g, \lambda) / \partial \lambda^2 - \mu \partial^2 C(g, \lambda) / \partial \lambda^2 = \frac{22500\alpha_1 30g^2}{(600+(1-\lambda)150g)^2} - \frac{\alpha_2 180}{\lambda^2} + \\ &\frac{22500\alpha_3 180g^2}{(1+(1-\lambda)150g)^2} \\ &= \frac{(9000)30(0.325\,79)^2}{(600+(1-0.493\,63)150(0.325\,79))^2} - \frac{18}{0.493\,63^2} + \frac{(2250)180(0.325\,79)^2}{(1+(1-0.493\,63)150(0.325\,79))^2} = -8.944\,3. \\ &\left| \begin{array}{ccc} 0 & -1520.4 & -18.934 \\ -1520.4 & 330.45 & 200.21 \\ -18.934 & 200.21 & -8.944\,3 \end{array} \right| = 3.208\,4 \times 10^7 > 0, \text{ so this is a maximum point.} \end{aligned}$$

Welfare in Example C

Without poverty alleviation, in this case, $g = \frac{\alpha_2 m^h}{\alpha_2 + \alpha_1 n^h} = \frac{10}{0.1+60} = 0.16639$ and welfare is given by

$$\begin{aligned} & n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + (n^h + n^l) \alpha_2 \ln \lambda n^h g^h \\ & + (n^h + n^l) \alpha_3 \ln(1 + (\frac{1-\lambda}{\lambda}) \lambda n^h g^h) \\ & = 60 \ln(100 - 0.16639) + 12 \ln(20) + 18 \ln(150(0.16639)) + 18 \ln(1) = 370.07 \end{aligned}$$

With poverty alleviation, with $g = 0.32579$, $\lambda = 0.49363$, welfare is given by

$$\begin{aligned} & 60 \ln(100 - 0.32579) + 12 \ln(20 + (1 - 0.49363) 5(0.32579)) + 18 \ln((0.49363) 150(0.32579)) + \\ & 18 \ln(1 + (1 - 0.49363) 150(0.32579)) = 428.31 > 370.07. \end{aligned}$$

In this case, the transfer to each poor household is $5(0.32579)(1 - 0.49363) = 0.82485$.

As a proportion of original income this is $0.82485/20 = 4.1243 \times 10^{-2}$. I.e., approximately 4%.

Example D.

Consider a case with $\alpha_3 > \alpha_2$.

$\alpha_1 = 0.4; \alpha_2 = 0.1; \alpha_3 = 0.2 \quad m^h = 100; m^l = 20 \quad n^h = 150; n^l = 30$ The system of FOC in this case is

$$\frac{(1-\lambda)12}{(4+(1-\lambda)g)} - \frac{60}{(100-g)} + \frac{18}{g} + \frac{36(1-\lambda)150}{(1+(1-\lambda)150g)} = \mu((1-\lambda)(4500 - 18090g) - 60.1) \quad (52)$$

$$-\frac{12g}{(4+(1-\lambda)g)} + \frac{18}{\lambda} - \frac{(36)150g}{(1+(1-\lambda)150g)} = \mu(-4500g + 9045g^2) \quad (53)$$

$$10 + 4500(1-\lambda)g - (60.1)g - 9045(1-\lambda)g^2 = 0 \quad (54)$$

$$\text{The solution is: } \left\{ \begin{array}{l} [g = 0.490\,98, \lambda = 0.327\,78, \mu = -3.652\,9 \times 10^{-2}], \\ [g = -21.815, \lambda = 0.999\,70, \mu = 1.477\,7], \\ [g = -0.533\,11, \lambda = 0.991\,54, \mu = 1.794\,4], \\ [g = 0.498\,19, \lambda = 7.515\,9, \mu = 1.434\,8 \times 10^{-4}] \end{array} \right\}$$

The only sensible real solution is $g = 0.490\,98, \lambda = 0.327\,78, \mu = -0.3652\,9$.

The Second-Order Condition for Example D

In this example,

$$\begin{aligned} \psi_1 &= \partial C(g, \lambda) / \partial g = (\alpha_2 + \alpha_3)(1 - \lambda)15000 - (\alpha_2 + \alpha_1)150 - (1 - \lambda)300(\alpha_2 + \alpha_3 + 150\alpha_1)g \\ &= (0.3)(1 - 0.327\,78)15000 - (0.1 + 60) - (1 - 0.327\,78)300(0.3 + 60)(0.490\,98) = -3005. \end{aligned}$$

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$$\begin{aligned} \psi_2 &= \partial C(g, \lambda) / \partial \lambda = -(\alpha_2 + \alpha_3)15000g + 150(\alpha_2 + \alpha_3 + 150\alpha_1)g^2 = \\ &= \partial C(g, \lambda) / \partial \lambda = -(0.3)15000(0.490\,98) + 150(0.3 + 60)(0.490\,98)^2 = -29.010 \end{aligned}$$

$$\begin{aligned} \psi_{11} &= \partial^2 L / \partial g^2 = -\frac{60}{(100-g)^2} - \frac{12(1-\lambda)^2 22500}{(600+(1-\lambda)150(g))^2} - \frac{18}{g^2} - \frac{36(1-\lambda)^2 22500}{(1+(1-\lambda)150g)^2} - \mu 300(1-\lambda)(0.3+60) \\ &= -\frac{60}{(100-(0.490\,98))^2} - \frac{12(1-0.327\,78)^2 22500}{(600+(1-0.327\,78)150(0.490\,98))^2} - \frac{18}{(0.490\,98)^2} - \frac{36(1-0.327\,78)^2 22500}{(1+(1-0.327\,78)150(0.490\,98))^2} - \\ &(-3.652\,9 \times 10^{-2})300(1-0.327\,78)(0.3+60) \\ &= 225.76 \end{aligned}$$

$$\begin{aligned} \psi_{12} &= \partial^2 L / \partial g \partial \lambda = -\frac{4500\alpha_1}{(600+(1-\lambda)150g)} + \frac{22500\alpha_1(1-\lambda)30g}{(600+(1-\lambda)150g)^2} - \frac{180\alpha_3 150}{(1+(1-\lambda)150g)} + \frac{180\alpha_3(1-\lambda)g 22500}{(1+(1-\lambda)150g)^2} - \\ &\mu(150(\alpha_2 + \alpha_3 + 150\alpha_1)2g - (\alpha_2 + \alpha_3)15000) \\ &= -\frac{1800}{(600+(1-0.327\,78)150(0.490\,98))} + \frac{9000(1-0.327\,78)30(0.490\,98)}{(600+(1-0.327\,78)150(0.490\,98))^2} - \frac{(36)150}{(1+(1-0.327\,78)150(0.490\,98))} \\ &+ \frac{36(1-0.327\,78)(0.490\,98)22500}{(1+(1-0.327\,78)150(0.490\,98))^2} - (-3.652\,9 \times 10^{-2})(150(0.3+60)2(0.490\,98) - (0.3)15000) = \\ &155.39 \end{aligned}$$

$$\begin{aligned} \psi_{22} &= \partial^2 L / \partial \lambda^2 = \partial^2 M(g, \lambda) / \partial \lambda^2 - \mu \partial^2 C(g, \lambda) / \partial \lambda^2 = \frac{22500\alpha_1 30g^2}{(600+(1-\lambda)150g)^2} - \frac{\alpha_2 180}{\lambda^2} + \\ &\frac{22500\alpha_3 180g^2}{(1+(1-\lambda)150g)^2} \\ &= \frac{(9000)30(\mathbf{0.490\,98})^2}{(600+(1-\mathbf{0.327\,78})150(\mathbf{0.490\,98}))^2} - \frac{18}{\mathbf{0.327\,78}^2} + \frac{(4500)180(\mathbf{0.490\,98})^2}{(1+(1-\mathbf{0.327\,78})150(\mathbf{0.490\,98}))^2} = -90.838 \end{aligned}$$

Therefore, the determinant of the bordered Hessian matrix is

$$\begin{vmatrix} 0 & -3005.7 & -29.010 \\ -3005.7 & 225.76 & 155.39 \\ -29.010 & 155.39 & -90.838 \end{vmatrix} = 8.4756 \times 10^8 > 0. \text{ So, this solution identifies a maximum.}$$

Welfare in Case D

$$\begin{aligned} \text{If } g = 0.49098, \lambda = 0.32778, \text{ welfare} &= n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + \\ & (n^h + n^l) \alpha_2 \ln \lambda n^h g^h + (n^h + n^l) \alpha_3 \ln(1 + (1 - \lambda) n^h g^h) \\ &= 60 \ln(100 - g^h) + 12 \ln(20 + (\frac{1-\lambda}{n^l}) 150 g^h) + 18 \ln \lambda 150 g^h + 18 \ln(1 + (1 - \lambda) 150 g^h) \\ &= 60 \ln(100 - 0.49098) + 12 \ln(20 + (1 - 0.32778) 5 (0.49098)) + 18 \ln((0.32778) 150 (0.49098)) + \\ & 18 \ln(1 + (1 - 0.32778) 150 (0.49098)) = 440.82 \end{aligned}$$

When there is no poverty alleviation and $\lambda = 1$, $g = \frac{10}{60.1} = 0.16639$ and $G = \frac{1500}{60.1} = 24.958$.

$$\text{Then welfare} = 60 \ln(100 - 0.16639) + 12 \ln 20 + 18 \ln 24.958 + 18 \ln(1) = 370.07$$

This is less than the welfare when poverty alleviation occurs.

The transfer to a poor household when $g = 0.49098, \lambda = 0.32778$ is $5(0.49098)(1 - 0.32778) = 1.6502$. so, as a proportion of the original total income of the poor, the transfer is $1.6502/20 = 0.08251$ - i.e., over 8%.

Example E.

$\alpha_1 = 0.45; \alpha_2 = 0.225; \alpha_3 = 0.225; m^h = 100; m^l = 20; n^h = 150, n^l = 30$. The charity's maximand is

$$\begin{aligned} M(g, \lambda) &= n^h [\alpha_1 \ln(m^h - g^h) + \alpha_2 \ln \lambda n^h g^h + \alpha_3 \ln(1 + (\frac{1-\lambda}{\lambda}) \lambda n^h g^h)] \\ &+ n^l [\alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + \alpha_2 \ln \lambda n^h g^h + \alpha_3 \ln(1 + (1 - \lambda) n^h g^h)] \\ &= n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + (n^h + n^l) \alpha_2 \ln \lambda n^h g^h \\ &+ (n^h + n^l) \alpha_3 \ln(1 + (1 - \lambda) n^h g^h) \\ &= 67.5 \ln(100 - g) + 40.5 (\ln 150 + \ln \lambda + \ln g) + 40.5 \ln(1 + 150(1 - \lambda)g) + 13.5 \ln(20 + (1 - \lambda)5g), \end{aligned}$$

substituting in parameters and suppressing the script in g^h . So,

$$\frac{\partial M(g, \lambda)}{\partial g} = -\frac{67.5}{100-g} + \frac{40.5}{g} + 40.5 \frac{150(1-\lambda)}{1+150(1-\lambda)g} + \frac{67.5(1-\lambda)}{20+5(1-\lambda)g}$$

$$\frac{\partial M(g, \lambda)}{\partial \lambda} = \frac{40.5}{\lambda} - 40.5 \frac{150g}{1+150(1-\lambda)g} - \frac{67.5g}{20+5(1-\lambda)g}$$

Likewise, substituting the parameters into the constraint and simplifying,

$$C(g, \lambda) = 0.225m^h + (0.45)(1-\lambda)m^h n^h g - (0.225 + 0.45n^h)g - n^h(0.45 + 0.45n^h)(1-\lambda)g^2 =$$

0

$$= 22.5 + (1-\lambda)(6750g - 10193g^2) - (67.725)g^h$$

$$\frac{\partial C(g, \lambda)}{\partial g} = (1-\lambda)(6750 - 20386g) - 67.725$$

$$\frac{\partial C(g, \lambda)}{\partial \lambda} = -6750g + 10193g^2$$

The system of FOC to be solved is

$$\frac{40.5}{g} - \frac{67.5}{100-g} + \frac{6075(1-\lambda)}{1+150(1-\lambda)g} + \frac{13.5(1-\lambda)}{4+(1-\lambda)g} = \mu((1-\lambda)(6750 - 20386g) - 67.725)$$
(55)

$$\frac{40.5}{\lambda} - \frac{6075g}{1+150(1-\lambda)g} - \frac{13.5g}{4+(1-\lambda)g} = -\mu(6750g + 10193g^2)$$
(56)

$$22.5 + (1-\lambda)(6750g - 10193g^2) - (67.725)g = 0$$
(57)

The solution is: $\left\{ \begin{array}{l} [g = 99.088, \lambda = 1.0001, \mu = -324.01], [g = 0.66287, \lambda = 6.1081, \mu = 2.7067 \times 10^{-4}] \\ [g = 0.65574, \lambda = 0.49374, \mu = -3.6045 \times 10^{-2}], \\ [g = 0.45636 - 4.9726i, \lambda = 1.0 + 1.3294 \times 10^{-3}i, \mu = 1.8036 - 9.1503 \times 10^{-2}i], \\ [g = 0.45636 + 4.9726i, \lambda = 1.0 - 1.3294 \times 10^{-3}i, \mu = 1.8036 + 9.1503 \times 10^{-2}i] \end{array} \right.$

The only feasible real solution now is $[g = 0.65574, \lambda = 0.49374, \mu = -.036045]$.

Checking the Second-order Condition in Example E

Using $[g = 0.65574, \lambda = 0.49374, \mu = -.036045]$,

$$\psi_1 = \partial C(g, \lambda) / \partial g = (\alpha_2 + \alpha_3)(1-\lambda)15000 - (\alpha_2 + \alpha_1 150) - (1-\lambda)300(\alpha_2 + \alpha_3 + 150\alpha_1)g$$

$$= (0.45)(1 - 0.49374)15000 - (0.225 + (0.45)150) - (1 - 0.49374)300(0.45 + 150(0.45))0.65574 =$$

$$-3417.8$$

$$\psi_2 = \partial C(g, \lambda) / \partial \lambda = -(\alpha_2 + \alpha_3) 15000g + 150(\alpha_2 + \alpha_3 + 150\alpha_1) g^2 =$$

$$= -(0.45) 15000 (0.65574) + 150(0.45 + 150(0.45)) (0.65574)^2 = -43.508$$

$$\psi_{11} = \partial^2 L / \partial g^2 = -\frac{150\alpha_1}{(100-g)^2} - \frac{30\alpha_1(1-\lambda)^2 22500}{(600+(1-\lambda)150(g))^2} - \frac{\alpha_2 180}{g^2} - \frac{180\alpha_3(1-\lambda)^2 22500}{(1+(1-\lambda)150g)^2} - \mu 300(1-\lambda)(\alpha_2 + \alpha_3 + 150\alpha_1)$$

$$= -\frac{150(0.45)}{(100-0.65574)^2} - \frac{30(0.45)(1-0.49374)^2 22500}{(600+(1-0.49374)150(0.65574))^2} - \frac{(0.225)180}{0.65574^2} - \frac{180(0.225)(1-0.49374)^2 22500}{(1+(1-0.49374)150(0.65574))^2} -$$

$$(-.036045) 300(1-0.49374)(0.45+150(0.45)) = 187.09$$

$$\psi_{12} = \partial^2 L / \partial g \partial \lambda = -\frac{4500\alpha_1}{(600+(1-\lambda)150(g))} + \frac{22500\alpha_1(1-\lambda)30g}{(600+(1-\lambda)150g)^2} - \frac{180\alpha_3 150}{(1+(1-\lambda)150g)} + \frac{180\alpha_3(1-\lambda)g 22500}{(1+(1-\lambda)150g)^2} -$$

$$\mu(150(\alpha_2 + \alpha_3 + 150\alpha_1) 2g - (\alpha_2 + \alpha_3) 15000)$$

$$= -\frac{4500(0.45)}{(600+(1-0.49374)150(0.65574))} + \frac{22500(0.45)(1-0.49374)30(0.65574)}{(600+(1-0.49374)150(0.65574))^2} - \frac{180(0.225)150}{(1+(1-0.49374)150(0.65574))}$$

$$+ \frac{180(0.225)(1-0.49374)(0.65574)22500}{(1+(1-0.49374)150(0.65574))^2} - (-.036045)(150(0.45+150(0.45))2(0.65574) - (0.45)15000)$$

$$= 233.29$$

$$\psi_{22} = \partial^2 L / \partial \lambda^2 = \partial^2 M(g, \lambda) / \partial \lambda^2 - \mu \partial^2 C(g, \lambda) / \partial \lambda^2 = \frac{22500\alpha_1 30g^2}{(600+(1-\lambda)150g)^2} - \frac{\alpha_2 180}{\lambda^2} +$$

$$\frac{22500\alpha_3 180g^2}{(1+(1-\lambda)150g)^2}$$

$$= \frac{22500(0.45)30(0.65574)^2}{(600+(1-0.49374)150(0.65574))^2} - \frac{(0.225)180}{0.49374^2} + \frac{22500(0.225)180(0.65574)^2}{(1+(1-0.49374)150(0.65574))^2} = -13.967$$

$$\begin{vmatrix} 0 & -3417.8 & -43.508 \\ -3417.8 & 187.09 & 233.29 \\ -43.508 & 233.29 & -13.967 \end{vmatrix} = 2.3218 \times 10^8 > 0$$

So, $[g = 0.65574, \lambda = 0.49374, \mu = -.036045]$ is a maximum point.

Welfare in Example E

Welfare with poverty alleviation is now

$$M(g, \lambda) = 67.5 \ln(100 - g) + 40.5(\ln 150 + \ln \lambda + \ln g) + 40.5 \ln(1 + 150(1 - \lambda)g) +$$

$$13.5 \ln(20 + (1 - \lambda)5g)$$

$$= 67.5 \ln(100 - 0.65574) + 40.5(\ln 150 + \ln 0.49374 + \ln 0.65574) +$$

$$40.5 \ln(1 + 150(0.65574) - 150(0.49374)0.65574) + 13.5 \ln(20 + (1 - 0.49374)5(0.65574)) =$$

$$= 67.5 \ln(99.34426) + 40.5(\ln 48.56476014) + 40.5 \ln(50.79623986) + 13.5 \ln(20 + 1.659874662) =$$

$$668.26$$

When $\lambda = 1$ and there is no poverty alleviation, with these parameters, $g = \frac{\alpha_2 m^h}{\alpha_2 + \alpha_1 n^h} = \frac{22.5}{0.225 + 67.5} = 0.332\ 23$. Thus, welfare is

$$\begin{aligned} & n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l) + (n^h + n^l) \alpha_2 \ln \lambda n^h g^h + (n^h + n^l) \alpha_3 \ln(1) \\ & = 150(0.45) \ln(100 - 0.3322259) + 30(0.45) \ln 20 + 180(0.225) \ln(150(0.33223)) = 509.37 < 668.26. \end{aligned}$$

So, poverty alleviation results in a significant increase in the total amount donated and in welfare

The transfer to each poor household in this case is $5(0.65574)(1 - 0.49374) = 1.6599$. As a proportion of a poor household's income, this is $1.6599/20 = 8.2995 \times 10^{-2}$, or approximately 8.3%.

Example F

$\alpha_1 = 0.45; \alpha_2 = 0.225; \alpha_3 = 0.21 \quad m^h = 100; m^l = 20 \quad n^h = 150; n^l = 30$. Proceeding as before yields the following FOC system to solve for the optimum:

$$-\frac{67.5}{(100 - g)} + \frac{(1 - \lambda) 13.5}{(4 + (1 - \lambda) g)} + \frac{40.5}{g} + \frac{(1 - \lambda) 5670}{(1 + (1 - \lambda) 150g)} = \mu((1 - \lambda)(6525 - 20380.5g) - 67.725) \quad (58)$$

$$-\frac{13.5}{(4 + (1 - \lambda) g)} + \frac{40.5}{\lambda} - \frac{5670g}{(1 + (1 - \lambda) 150g)} = \mu(-6525g + 150(67.935)g^2) \quad (59)$$

$$22.5 + (1 - \lambda)(6525g - 10190.25g^2) - (67.725)g = 0 \quad (60)$$

The solution is:

$$\left\{ \begin{array}{l} [g = 0.633\,90, \lambda = 0.507\,3, \mu = -3.826\,4 \times 10^{-2}], \\ \left[\begin{array}{l} g = -1.4199 \times 10^{-4} - 9.860\,6 \times 10^{-4}i, \lambda = 0.501\,68 + 3.426\,0i, \\ \mu = 1.288\,2 \times 10^{-3} + 5.292\,7 \times 10^{-3}i \\ g = -1.4199 \times 10^{-4} + 9.860\,6 \times 10^{-4}i, \lambda = 0.501\,68 - 3.426\,0i, \\ \mu = 1.288\,2 \times 10^{-3} - 5.292\,7 \times 10^{-3}i \end{array} \right], \\ [g = 0.640\,98, \lambda = 5.818\,8, \mu = 1.597\,2 \times 10^{-3}], \\ [g = 1.673\,2 - 4.630\,0i, \lambda = 1.000\,4 + 1.316\,7 \times 10^{-3}i, \mu = 1.826\,4 - 8.592\,5 \times 10^{-2}i], \\ [g = 1.673\,2 + 4.630\,0i, \lambda = 1.000\,4 - 1.316\,7 \times 10^{-3}i, \mu = 1.826\,4 + 8.592\,5 \times 10^{-2}i] \end{array} \right\}$$

The only real feasible solution is $g = 0.633\,90, \lambda = 0.507\,3, \mu = -.03\,826\,4$

The Second-order condition in Example F

We proceed as before to get

$$\begin{aligned} \psi_1 &= \partial C(g, \lambda) / \partial g = (\alpha_2 + \alpha_3)(1 - \lambda)15000 - (\alpha_2 + \alpha_1 150) - (1 - \lambda)300(\alpha_2 + \alpha_3 + 150\alpha_1)g \\ &= (0.435)(1 - 0.507\,3)15000 - (0.225 + 67.5) - (1 - 0.507\,3)300(0.435 + 67.5)(0.633\,90) = \\ &-3218.1 \end{aligned}$$

$$\begin{aligned} \psi_2 &= \partial C(g, \lambda) / \partial \lambda = -(\alpha_2 + \alpha_3)15000g + 150(\alpha_2 + \alpha_3 + 150\alpha_1)g^2 \\ &= -(0.435)15000(0.633\,90) + 150(0.435 + 67.5)(0.633\,90)^2 = -41.457 \end{aligned}$$

$$\begin{aligned} \psi_{11} &= \partial^2 L / \partial g^2 = -\frac{150\alpha_1}{(100-g)^2} - \frac{30\alpha_1(1-\lambda)^2 22500}{(600+(1-\lambda)150(g))^2} - \frac{\alpha_2 180}{g^2} - \frac{180\alpha_3(1-\lambda)^2 22500}{(1+(1-\lambda)150(\mathbf{g}))^2} - \mu 300(1-\lambda)(\alpha_2 + \alpha_3 + 150\alpha_1) \\ &= -\frac{67.5}{(100-(0.633\,90))^2} - \frac{13.5(1-0.507\,3)^2 22500}{(600+(1-0.507\,3)150(0.633\,90))^2} - \frac{40.5}{(0.633\,90)^2} - \frac{37.8(1-0.507\,3)^2 22500}{(1+(1-0.507\,3)150(0.633\,90))^2} - \\ &(-.03\,826\,4)300(1-0.507\,3)(0.435+67.5) = 193.08 \end{aligned}$$

$$\begin{aligned} \psi_{12} &= \partial^2 L / \partial g \partial \lambda = -\frac{4500\alpha_1}{(600+(1-\lambda)150g)} + \frac{22500\alpha_1(1-\lambda)30g}{(600+(1-\lambda)150g)^2} - \frac{180\alpha_3 150}{(1+(1-\lambda)150g)} + \frac{180\alpha_3(1-\lambda)g 22500}{(1+(1-\lambda)150g)^2} - \\ &(\mu)(150(\alpha_2 + \alpha_3 + 150\alpha_1)2g - (\alpha_2 + \alpha_3)15000) \\ &= -\frac{2025}{(600+(1-0.507\,3)150(0.633\,90))} + \frac{10125(1-0.507\,3)30(0.633\,90)}{(600+(1-0.507\,3)150(0.633\,90))^2} - \frac{(37.8)150}{(1+(1-0.507\,3)150(0.633\,90))} + \\ &\frac{37.8(1-0.507\,3)(0.633\,90)22500}{(1+(1-0.507\,3)150(0.633\,90))^2} - (-.03\,826\,4)(150(.435+67.5)2(0.633\,90) - (0.435)15000) = \end{aligned}$$

239.29

$$\psi_{22} = \partial^2 L / \partial \lambda^2 = \partial^2 M(g, \lambda) / \partial \lambda^2 - \mu \partial^2 C(g, \lambda) / \partial \lambda^2 = \frac{22500\alpha_1 30g^2}{(600+(1-\lambda)150g)^2} - \frac{\alpha_2 180}{\lambda^2} +$$

$$\frac{22500\alpha_3 180g^2}{(1+(1-\lambda)150g)^2} = \frac{(10125)30(0.63390)^2}{(600+(1-0.5073)150(0.63390))^2} - \frac{40.5}{0.5073^2} + \frac{(4725)180(0.63390)^2}{(1+(1-0.5073)150(0.63390))^2} = -7.8065$$

Hence, the determinant of the bordered Hessian matrix is

$$\begin{vmatrix} 0 & -3218.1 & -41.457 \\ -3218.1 & 193.08 & 239.29 \\ -41.457 & 239.29 & -7.8065 \end{vmatrix} = 1.4436 \times 10^8 > 0.$$

So $g = 0.63390$, $\lambda = 0.5073$, $\mu = -0.038264$ is a maximum point.

Welfare in Example F

$$\begin{aligned} \text{With } g = 0.63390, \lambda = 0.5073, \text{ welfare is given by } & n^h \alpha_1 \ln(m^h - g^h) + n^l \alpha_1 \ln(m^l + (\frac{1-\lambda}{n^l}) n^h g^h) + \\ & (n^h + n^l) \alpha_2 \ln \lambda n^h g^h + (n^h + n^l) \alpha_3 \ln(1 + (\frac{1-\lambda}{\lambda}) \lambda n^h g^h) \\ & = 67.5 \ln(m^h - g^h) + 13.5 \ln(20 + (1-\lambda) 5g^h) + 40.5 \ln \lambda 150g^h + 37.8 \ln(1 + (1-\lambda) 150g^h) \\ & = 67.5 \ln(100 - 0.63390) + 13.5 \ln(20 + (1-0.5073) 5(0.63390)) + 40.5 \ln((0.5073) 150(0.63390)) + \\ & 37.8 \ln(1 + (1-0.5073) 150(0.63390)) = 655.07 \end{aligned}$$

Without poverty alleviation, giving satisfies $g = \frac{\alpha_2 m^h}{\alpha_2 + \alpha_1 n^h} = \frac{22.5}{0.225 + 67.5} = 0.33223$. Then welfare = $67.5 \ln(100 - 0.33223) + 13.5 \ln(20) + 40.5 (\ln(150(0.33223))) + 37.8 \ln(1) = 509$.

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The poverty alleviating transfer in this case to each poor person = $5(0.63390)(1 - 0.5073) = 1.5616$. As a proportion of original income, this equals

$$1.5616/20 = 0.07808 \text{ - i.e., nearly 8\%}.$$

References

Murata, Y. 1977. *Mathematics for Stability and Optimization of Economic Systems*. London: Academic Press.